

The Multifaceted Role of EXOC6A in Ciliogenesis

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eLife Assessment

This **important** study elucidates the role of the exocyst component EXOC6A at distinct stages of ciliogenesis, which advances our understanding of ciliary membrane remodeling and cilium formation. The authors provide **solid** evidence that EXOC6A interacts with myosin-Va and is dynamically recruited via dynein-, microtubule-, and actin-dependent mechanisms, to support proper formation of the ciliary membrane. The study will be of interest to cell biologists and other researchers interested in vesicular trafficking, organellar membrane dynamics, and ciliogenesis.

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Abstract

Ciliogenesis is a highly ordered process that requires membrane trafficking, fusion, and maturation. In this study, we investigated EXOC6A, a component of the exocyst complex known for secretory vesicle trafficking and fusion, and found that it interacts with myosin-Va (Myo-Va) during ciliogenesis. EXOC6A colocalizes with Myo-Va at various stages of ciliogenesis, including preciliary vesicles, ciliary vesicles (CVs), and ciliary sheath membrane during ciliogenesis. We found that EXOC6A vesicles are actively recruited, integrated, and exit from the CVs and the ciliary sheath, implying that EXOC6A vesicles may facilitate continuous cilia membrane remodeling during ciliogenesis. Importantly, EXOC6A knockout impairs ciliogenesis, arresting most cells at the CV stage and preventing recruitment of NPHP and MKS module components to the transition zone. Furthermore, EXOC6A vesicles are transported to the mother centriole via a dynein-, microtubule-, and actin-dependent mechanism. Our results suggest that EXOC6A functions in both early and late stages of ciliogenesis and is involved in orchestrating vesicle dynamics, cilia membrane remodeling, and formation.

Introduction

Most mammalian cells build primary cilia, which act as chemosensors or mechanosensors and as transducers that regulate key developmental signaling pathways. Primary cilia are composed of nine doublet microtubules arranged in a ring and surrounded by a ciliary membrane that is continuous with the plasma membrane (Dawe et al., 2007 [↗](#); Nigg and Raff, 2009 [↗](#)). Primary cilia protrude from the cell surface to perform a wide range of biological functions. Defects in ciliogenesis can lead to a number of genetic disorders known as ciliopathies, which are characterized by loss of vision, disturbed kidney function, obesity, and defects in brain development (Sattar and Gleeson, 2011 [↗](#); Valente et al., 2014 [↗](#)).

Ciliogenesis is a highly ordered and complex process. Currently, two distinct pathways have been reported, with different cell types involved: intracellular and extracellular. Polarized epithelial cells use the extracellular pathway, whereas fibroblasts and mesenchymal cells use the intracellular pathway (Breslow and Holland, 2019 [↗](#)). In the intracellular pathway, myosin-Va (Myo-Va) facilitates the transportation of preciliary vesicles (PCVs) to the distal appendages (DAs) of the M-centrioles (Wu et al., 2018 [↗](#)). These PCVs may be derived from the Golgi apparatus (Schmidt et al., 2012 [↗](#)) and subsequently fuse with a ciliary vesicle (CV) through the action of membrane fusion regulators EHD1 and EHD3 (Lu et al., 2015 [↗](#)).

This step is critical for the conversion of the mother centriole into a basal body, which is responsible for removing the centriole cap protein, CP110, and recruiting ciliary transition zone (TZ) proteins (Lu et al., 2015 [↗](#)). As a result, an axoneme extends within the larger CV, which gradually stretches out to form the double membranes of the ciliary shaft and sheath that surround the extending axoneme. Finally, the ciliary sheath (the outer membrane) fuses with the plasma membrane and the axoneme, enveloped by the ciliary shaft membrane, extends from the cell membrane, and makes contact with the extracellular environment (Ghossoub et al., 2011 [↗](#)). A signaling cascade including RAB11, RABIN8, and RAB8 has also been reported to regulate the ciliary membrane assembly (Chiba et al., 2013 [↗](#); Feng et al., 2012 [↗](#); Knödler et al., 2010 [↗](#); Nachury et al., 2007 [↗](#); Westlake et al., 2011 [↗](#); Yoshimura et al., 2007 [↗](#)). However, the processes of formation of the ciliary sheath and ciliary shaft membrane during ciliogenesis remain unclear.

The exocyst is a multi-subunit protein complex that was first identified in yeast, which mediates the tethering of secretory vesicles to the plasma membrane (Mei and Guo, 2018 [↗](#)). The complex is composed of eight subunits, named Sec (for secretion; Sec3, Sec5, Sec6, Sec8, Sec10, Sec15, Exo70, and Exo84) in yeast or Exoc (for exocyst-related; EXOC1 to 8) in humans (TerBush et al., 1996 [↗](#); TerBush and Novick, 1995 [↗](#)). The exocyst complex is implicated in various cellular processes, including exocytosis, ciliogenesis, cytokinesis, autophagy, cell polarity, migration, tumorigenesis, and fusion of secretory vesicles (Mei and Guo, 2018 [↗](#)).

Mutations in genes encoding components of the exocyst complex have been associated with various human genetic disorders. For example, a mutation in *EXOC8* has been identified in Joubert syndrome (Dixon-Salazar et al., 2012 [↗](#)), while a mutation in *EXOC4* has been reported in Meckel-Gruber syndrome (Shaheen et al., 2013 [↗](#)). Both of these disorders involve clinically and genetically heterogeneous ciliopathy. Additionally, mutations in *EXOC2* cause severe defects in human brain development (Van Bergen et al., 2020 [↗](#)), while mutations in *EXOC7* and *EXOC8* result in a novel disorder of cerebral cortex development, characterized by brain atrophy, seizures, developmental delay, and microcephaly (Coulter et al., 2020 [↗](#)). However, the essential roles of these genes in ciliogenesis and their correlation with human genetic disorders are currently unknown.

It has been reported that some components of the exocyst complex are involved in ciliogenesis. For example, Sec10 (EXOC5) was shown to be required for ciliogenesis and cyst formation *in vitro*, and Sec10 knockdown leads to reduce ciliogenesis (Zuo et al., 2009 [↗](#); Zuo et al., 2019 [↗](#)). Furthermore, other subunits such as EXOC3 and EXOC4 have been detected at the cilium with unknown function (Seixas et al., 2016 [↗](#)). Interestingly, depletion of Sec8, Exo70, or Sec10 does not impair ciliogenesis (Rivera-Molina et al., 2021 [↗](#)), whereas knockdown of Sec15a (EXOC6A) significantly perturbs ciliogenesis (Rivera-Molina et al., 2021 [↗](#)). These findings imply that different exocyst subunits may have distinct, non-redundant functions during ciliogenesis. However, the precise localization of EXOC6A within cilia and its specific role in regulating ciliary membrane formation or trafficking events remain largely unknown.

Previously, we demonstrated that the earliest event in ciliogenesis is the transportation of PCVs by Myo-Va to the distal appendages of the M-centriole, and we showed that Myo-Va is released from the ciliary membrane upon fusion of the ciliary sheath with the plasma membrane (Wu et al., 2018 [↗](#)). However, the mechanism of the assembly of double ciliary membranes (ciliary sheath and shaft) during intracellular ciliogenesis was left unknown. In this study, we provide evidence that the exocyst component EXOC6A interacts with Myo-Va and plays an essential role in ciliary membrane assembly at different stages during ciliogenesis.

Results

Colocalization patterns of EXOC6A and Myo-Va at preciliary vesicles, ciliary vesicles, and ciliary sheath during intracellular ciliogenesis

To investigate the localization of EXOC6A during ciliogenesis, we treated retinal pigment epithelial (RPE1)-based mCherry-tagged Myo-Va cargo binding domain (Myo-Va-GTD)-inducible cells with doxycycline (Dox) for 24 h and then serum starved them for 2h to induce ciliogenesis. We used antibodies against endogenous EXOC6A and glutamylated tubulin (Glu-tub), a marker of centrioles and ciliary axoneme, for confocal immunofluorescence analysis. Our results showed that the EXOC6A signal was mainly detected as a dot-shaped structure resembling a ciliary vesicle (CV) at the distal end of the M-centriole (**Fig. 1A** [↗](#), left), while some EXOC6A signals appeared to be colocalized with mCherry-tagged Myo-Va-GTD at the ciliary membrane (**Fig. 1A** [↗](#), right). To determine the subcellular localization of EXOC6A, we performed immunofluorescence staining using antibodies against CEP164 (a distal appendage marker, (Graser et al., 2007 [↗](#))) and CEP120 (a centriole marker, (Lin et al., 2013 [↗](#))). The PCV/CV/ciliary sheath can be visualized with mCherry-tagged Myo-Va-GTD (Wu et al., 2018 [↗](#)). The cells were then analyzed using three-dimensional structured illumination microscopy (3D-SIM, **Fig. 1B** [↗](#)) or ultrastructure expansion microscopy (U-ExM, **Fig. 1C** [↗](#)). Our results showed that EXOC6A signals colocalized with mCherry-Myo-Va-GTD at CV (**Fig. 1B and C** [↗](#)). To further define the position of the EXOC6A signal, we performed correlative fluorescence light and electron microscopy (CFLEM) in RPE1-based GFP-EXOC6A-inducible cells (**Fig. 1D** [↗](#)). Our CFLEM results showed that the GFP-EXOC6A signals appeared to localize at PCVs (or vesicles), CVs (**Fig. 1D-i-ii** [↗](#)), and the elongating ciliary membrane (**Fig. 1D-iii-iv** [↗](#)).

To precisely analyze the subcellular localization of EXOC6A and Myo-Va during the formation of the ciliary sheath and the shaft membrane, we used 3D-SIM to analyze the RPE1 cells that exogenously express inducible GFP-EXOC6A using known markers combined for the ciliary sheath (Myo-Va-GTD; (Wu et al., 2018 [↗](#))), the ciliary shaft (ARL13B, INPP5E; (Feng et al., 2012 [↗](#); Garcia-Gonzalo et al., 2015 [↗](#); Wu et al., 2018 [↗](#))), and the axoneme (Glu-tub). EXOC6A signals were commonly detected at the Myo-Va-GTD-labeled ciliary sheath (**Fig. 1E-i** [↗](#), ii), which surrounds the ciliary shaft membrane (labeled ARL13B and INPP5E, **Fig. 1E-i** [↗](#), iii) and the axoneme (Glu-tub,

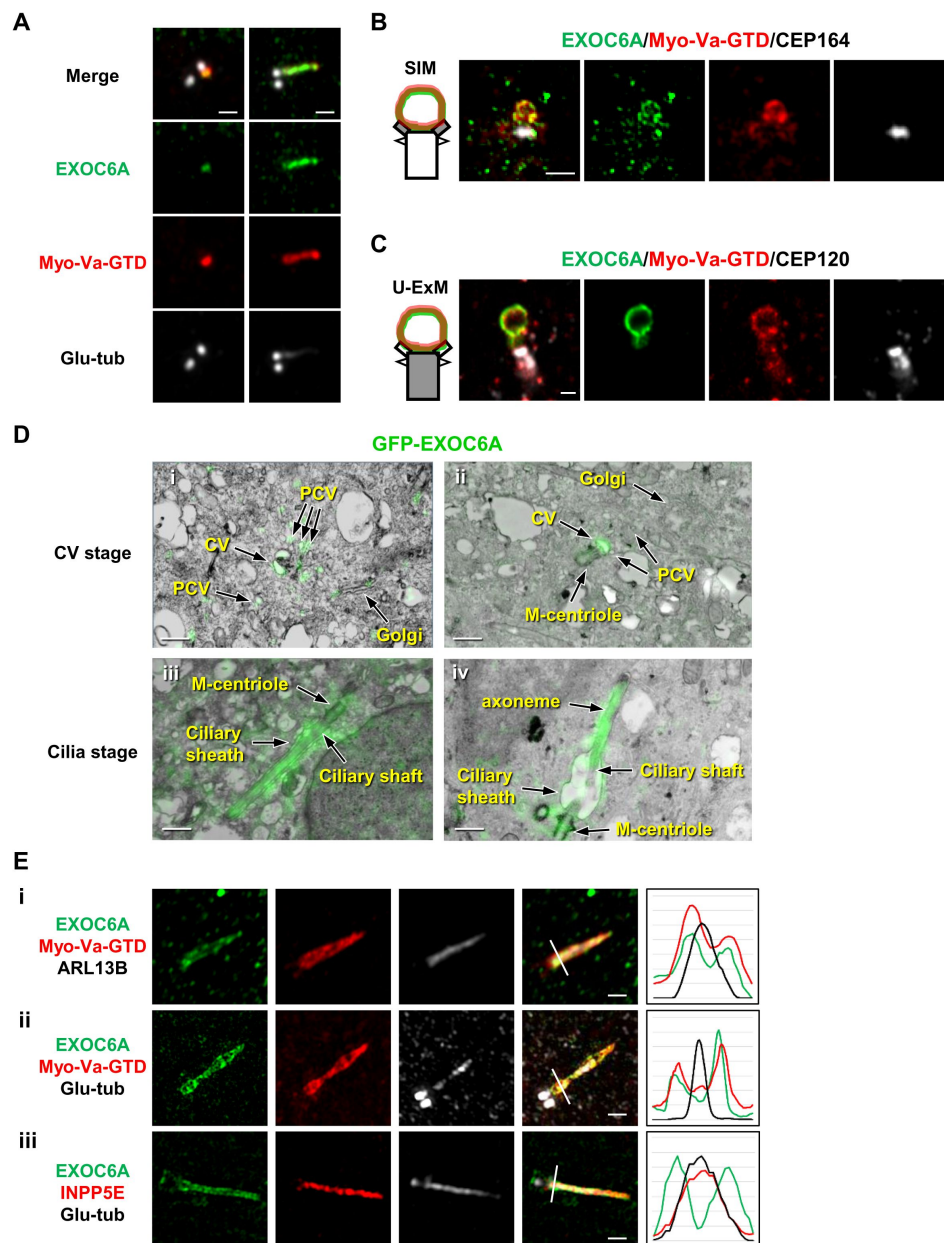


Figure 1.

Colocalization patterns of EXOC6A and myosin-Va at preciliary vesicles, ciliary vesicles, and ciliary sheath during ciliogenesis.

(A) RPE1-based mCherry-Myo-Va-GTD cells were treated with Dox for 24 h, followed by serum starvation for 2h. Cells were then stained with antibodies against EXOC6A (green) and polyglutamylated tubulin (Glu-tub) (gray) and analyzed using immunofluorescence confocal microscopy. (B and C) RPE1-based mCherry-Myo-Va-GTD-inducible cells were treated as described in A and immunostained with antibodies against EXOC6A (green), CEP120 (centriole marker), and CEP164 (distal appendage marker, gray). Images were captured using 3D-SIM or ultra-expansion microscopy (U-ExM) with an LSM880 confocal system. (D) CFLEM analysis of the localization of GFP-EXOC6A during ciliogenesis. RPE1-based GFP-EXOC6A-inducible cells were treated with Dox for 24 h and subjected to serum starvation for 2 or 24 h to observe the CVs or cilia membrane, respectively. Images were taken via SIM and TEM and then merged based on their relative localization. GFP-EXOC6A signals were located at PCVs, CVs (i-ii), and the ciliary membrane (iii-iv). (E) RPE1-based GFP-EXOC6A and mCherry-Myo-Va-GTD-inducible cells were treated with Dox for 24 h and analyzed via 3D-SIM using the indicated antibodies. Right panels show fluorescence profile plots. EXOC6A colocalized with myosin-Va at the ciliary sheath, while ARL13B (i) and INPP5E (iii) are the ciliary shaft markers. Scale bars are 1 μ m.

Fig. 1E-ii, iii [↗](#)). Since our 3D-SIM analysis could not distinguish the spatial distances between INPP5E and the Glu-tub-labeled axoneme due to the limits of its resolution power (100–130 nm), we thus conclude that EXOC6A localizes at the ciliary sheath membrane surrounding the ciliary shaft. Altogether, our data show that EXOC6A, like Myo-Va, localizes at the PCVs, the CVs, and the ciliary sheath membrane.

Spatial and temporal localization of Myo-Va and EXOC6A during ciliogenesis

To investigate the temporal and spatial correlation between the ciliary localization of EXOC6A and Myo-Va during ciliogenesis, we used RPE1-based mCherry-Myo-Va-GTD-inducible cells that were serum starved, released at different time points, and analyzed using 3D-SIM with antibodies against endogenous EXOC6A and Glu-tub (**Fig. 2A** [↗](#)). We identified five different phenotypes based on the immunostaining patterns of Myo-Va and EXOC6A during cilium assembly (**Fig. 2B** [↗](#)), which are quantified in **Fig. 2C** [↗](#). Before serum starvation, a number of EXOC6A-associated vesicles were detected close to the M-centriole (**Fig. 2A** [↗](#), 0 min, type 1). During the progression of serum starvation, EXOC6A-associated vesicles appeared to gradually accumulate near the M-centriole (**Fig. 2A** [↗](#), 15 min, type 2) in a process we called “clustering-to-centriole,” and subsequently, EXOC6A signals were mainly detected at the Myo-Va-labeled CVs (**Fig. 2A** [↗](#), 30 min–1 h, type 3) and the ciliary sheath membrane, surrounding the Glu-tub-labeled axoneme (**Fig. 2A** [↗](#), 2 h, type 4). In some cells, a tubule-like structure (EXOC6A⁺/Myo-Va⁺) linked to the ciliary sheath membrane was observed (**Fig. 2A** [↗](#), 4 h, white arrow). Finally, both EXOC6A and Myo-Va were released from the ciliary sheath membrane when it fused to the plasma membrane, while the Glu-tub-labeled axoneme continued to protrude into the extracellular milieu (**Fig. 2A** [↗](#), 24 h, type 5). Interestingly, we observed that only a portion, but not all of the EXOC6A-labeled vesicles colocalized with Myo-Va-labeled PCVs (**Fig. 2A** [↗](#), 0 and 15 min). However, once the CVs formed, both EXOC6A and Myo-Va-GTD signals were detected at the CVs (**Fig. 2A** [↗](#), 30 min, 45 min, and 1 h). We present a schematic model in **Figure 2B** [↗](#) that shows the localization of EXOC6A (green) and mCherry-Myo-Va-GTD (red) during ciliogenesis and the types of immunostaining patterns.

To further understand the relationship between the clustering-to-centriole progression of EXOC6A and Myo-Va during the early stage of ciliogenesis, we analyzed the intensity of both proteins clustering to the centrioles and calculated their overlapping co-efficiency correlation R (Pearson correlation coefficient) using Zeiss Zen software (**Fig. S1**). We observed a gradual increase in the mean intensity of EXOC6A and Myo-Va clustering to the centrioles from 0 to 30 min (**Fig. S1A–C**). Additionally, the correlation R between EXOC6A and Myo-Va significantly increased from 0 to 30 min (**Fig. S1D**). These results suggest a high correlation between Myo-Va-labeled vesicles and EXOC6A-labeled vesicles during the progression of clustering-to-centriole.

FRAP analysis of the dynamic localization of Myo-Va- and EXOC6A-labeled vesicles during ciliogenesis

We found that some EXOC6A labels had a tubule-like structure extending from the ciliary sheath (**Fig. 2A** [↗](#), 4 h, type 4), suggesting the dynamic assembly of the ciliary membrane during ciliogenesis. To gain further insights into the dynamic relationship between Myo-Va- and EXOC6A-labeled vesicles during early CV formation and the later sheath membrane assembly stage, we performed live cell imaging with fluorescence recovery after photobleaching (FRAP). FRAP is a widely used technique to measure the dynamics of ciliary membrane protein localization ([Kee et al., 2012](#) [↗](#); [Westlake et al., 2011](#) [↗](#)). After photobleaching at CVs or the ciliary sheath region, the EXOC6A signals were recurrently recruited to the CVs (**Fig. 3A** [↗](#), Movie 1) and the ciliary sheath membrane immediately (**Fig. 3B** [↗](#), Movie 2). Together, our data indicated that the EXOC6A-labeled vesicles are continuously recruited to and fused with the ciliary membrane not only at the CVs but also at the ciliary sheath. To further investigate the dynamic properties of the ciliary membrane, we performed live-cell super-resolution imaging using the ELYRA 7 SIM system. We

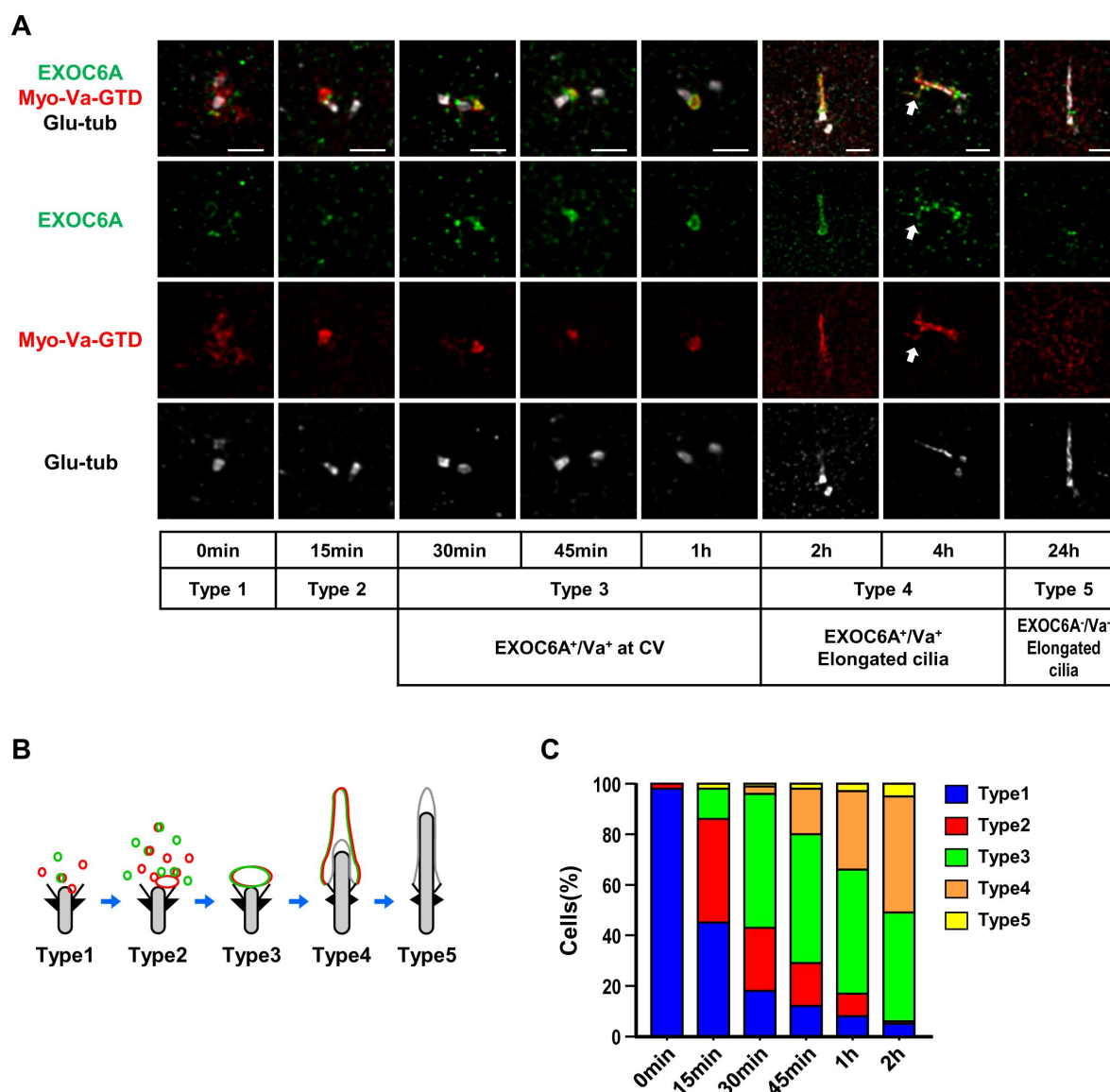


Figure 2.

Spatial-temporal localization of EXOC6A and Myo-Va during ciliogenesis.

(A) RPE1-based mCherry-Myo-Va-GTD-inducible cells were treated with Dox for 24 h, serum starved, released at different time points, and analyzed via 3D-SIM using the indicated antibodies. The white arrow indicates a tubule extending from the ciliary membrane. (B, C) Schematic model showing the localization patterns of EXOC6A (green) and Myo-Va (red) during ciliogenesis, and the percentages of cells with different types of staining patterns (%) are shown in C. Scale bars are 1 μ m.

used SiR-tubulin, a fluorescent probe derived from the microtubule-stabilizing drug docetaxel, to label endogenous microtubules. This allowed us to visualize the structure of centrioles and axonemes during ciliogenesis in live imaging experiments (Fig. S2). Our results show that EXOC6A co-localizes with Myo-Va-GTD on the ciliary membrane and surrounds the SiR-tubulin-labeled axoneme (Fig. S2A and S2B). In addition, we detected a dynamic tubule-like structure that extended and retracted from the ciliary membrane (Fig. S2A, S2C, and Movie 3). To better observe vesicle trafficking and membrane dynamics, we used the single-channel and burst modes of the ELYRA 7 SIM system to achieve higher temporal resolution in live imaging. Our results showed that GFP-EXOC6A-labeled vesicles gradually moved to and fused with CVs (Fig. 3C, white arrow; Movie 4) and the ciliary pocket, a membrane domain found at the base of primary cilia (Fig. 3D, white arrow, Movie 5). Interestingly, almost at the same time, some GFP-EXOC6A-labeled vesicles were fused with or secreted from the ciliary pocket (Fig. 3D, red arrow, Movie 5). Unexpectedly, we observed that some GFP-EXOC6A-associated tubular structures (Fig. 3D, yellow arrow) or GFP-EXOC6A-labeled vesicles were released directly from the ciliary membrane rather than from the ciliary pocket (Fig. 3D, orange arrow). Together, our results indicate that GFP-EXOC6A-associated vesicles/membranes are highly dynamic and continuously recruited to and/or released from CVs during the early stages of ciliogenesis, and that they are also fused with or excreted from the ciliary membrane during later stages of ciliogenesis. We, therefore, propose that EXOC6A vesicles are responsible for the exchange of materials within existing CVs and the ciliary sheath during ciliogenesis, facilitating the addition or removal of membranes or membrane proteins.

Myo-Va and EHD1 are required for EXOC6A-labeled CV formation, and EXOC6A is associated with Myo-Va

We previously reported that Myo-Va-mediated transportation of PCVs to the M-centriole is the earliest event that defines the onset of ciliogenesis (Wu et al., 2018). In this study, we aimed to examine the sequential order of PCV transportation mediated by Myo-Va and EXOC6A during ciliogenesis in both RPE1-based *Myo-Va* knockout (KO) and *EXOC6A* KO cells. Our results showed that few EXOC6A-labeled vesicles and no EXOC6A-labeled CVs were detected at the distal end of the M-centriole in the *Myo-Va* KO cells (Fig. 4A). In contrast, Myo-Va-labeled CVs were frequently observed at the M-centrioles of *EXOC6A* KO cells (Fig. 4B). Together, our findings suggest that EXOC6A is not essential for the formation of Myo-Va-labeled CVs during ciliogenesis.

EHD1 was reported to be essential for the fusion of PCVs to form larger CVs during the early stage of ciliogenesis (Lu et al., 2015). To investigate the sequential order of EHD1- and EXOC6A-mediated ciliary membrane assembly, we depleted EHD1 expression using siRNA in RPE1 cells. Our results showed that EHD1 depletion led to a reduction in the formation of EXOC6A-labeled CVs (Fig. 4C). In contrast, *EXOC6A* KO did not appear to affect EHD1-mediated fusion of PCVs to form a large EHD1-labeled CV at the M-centriole (Fig. 4D). Together, our findings suggest that EHD1 is required for EXOC6A-labeled CV formation, while EXOC6A is not essential for the formation of EHD1-labeled CVs.

Additionally, previous studies reported that Myo2p, the yeast homolog of human Myo-Va, interacts with sec15, the yeast homolog of human EXOC6A (Jin et al., 2011). The present study showed that some, but not all of the EXOC6A-labeled PCVs colocalized with Myo-Va-labeled PCVs (Fig. 2A, 0 and 15 min). We speculate that Myo-Va may transport some EXOC6A-labeled vesicles to the CVs for ciliary membrane assembly. Our co-immunoprecipitation experiments showed that endogenous EXOC6A can co-immunoprecipitate with Myo-Va, supporting this hypothesis (Fig. 4E).

Finally, previous studies have indicated that the removal of centriolar protein CP110 from the distal end of the M-centriole is necessary for axoneme growth (Spektor et al., 2007). It has been proposed that CP110 removal occurs after the docking of CV to the DA of the M-centriole (Lu et al., 2015; Wu et al., 2018). To investigate the impact of EXOC6A loss on CP110 removal, we

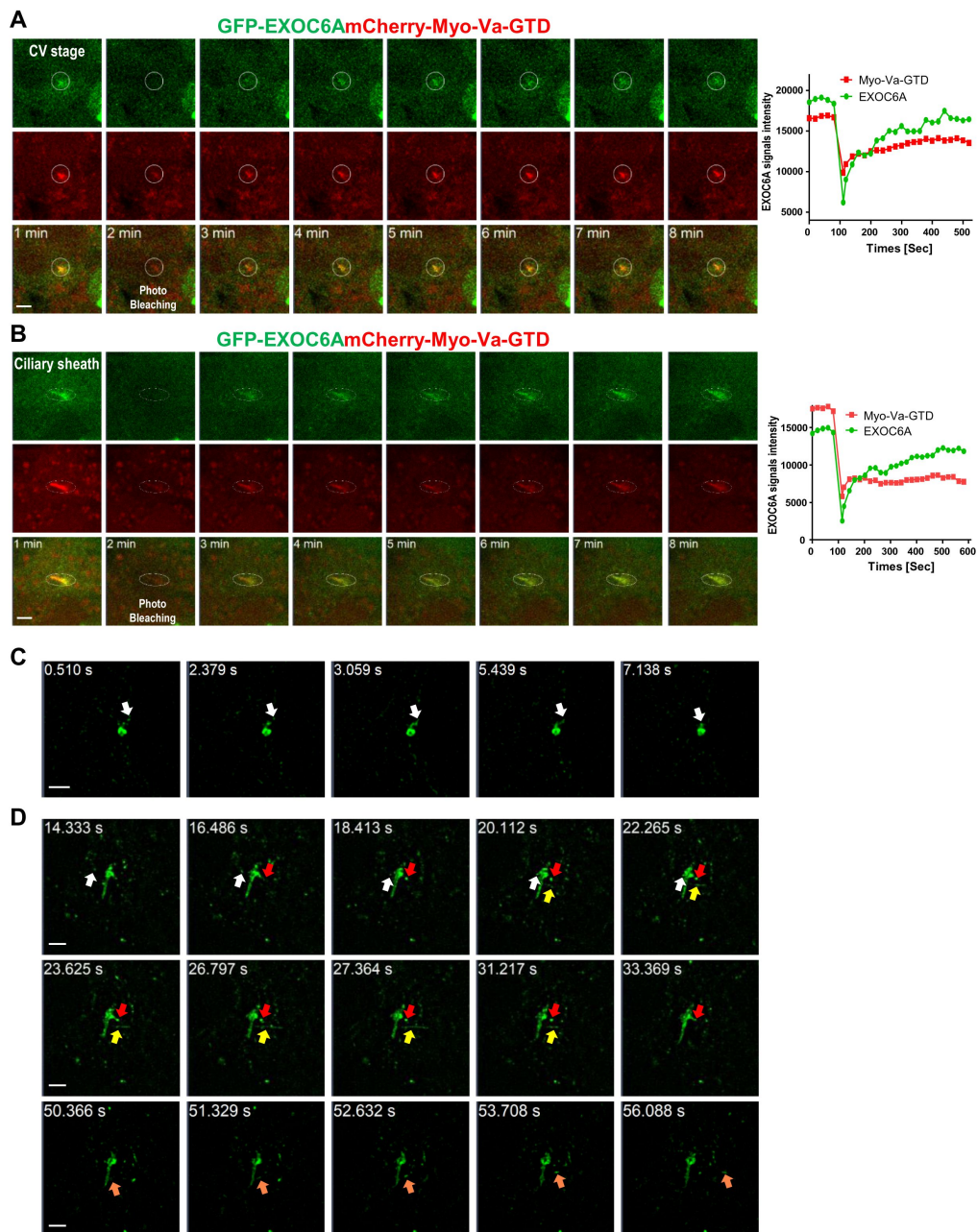


Figure 3.

FRAP analysis of Myo-Va- and EXOC6A-labeled vesicles during ciliogenesis.

(A and B) RPE1-based GFP-EXOC6A and mCherry-Myo-Va-GTD-inducible cells were treated with Dox for 24 h and serum starved for 30 min. Regions of CVs (A) or the ciliary membrane (B) of the cells were photobleached using the LSM880 confocal microscope with the 405 nm laser (related to Movies 1 and 2). Signal intensity of EXOC6A is shown in the right panel. (C and D) Dynamic localization of GFP-EXOC6A at CVs (C) or the ciliary membrane (D) during ciliogenesis. RPE1-based GFP-EXOC6A-inducible cells were treated with Dox for 24 h and serum starved for 30 min. Images were taken using an Elyra 7 high-speed live-cell imaging system. Series of images in (C) shows the progression of fusion of GFP-EXOC6A-associated vesicles into CVs (white arrow). Series of images in (D) shows the process by which GFP-EXOC6A-associated vesicles fuse into the ciliary pocket (white arrow) and simultaneously exit or reintegrate into the ciliary pocket (red arrow). Some images also show the release of tubule-like structures in the ciliary membrane (yellow arrow), or the excretion of vesicles in the ciliary membrane (orange arrow). These data suggested that the components of CVs, the ciliary pocket, and the ciliary membrane are highly dynamic (related to Movies 4 and 5). Scale bars are 2 μ m.

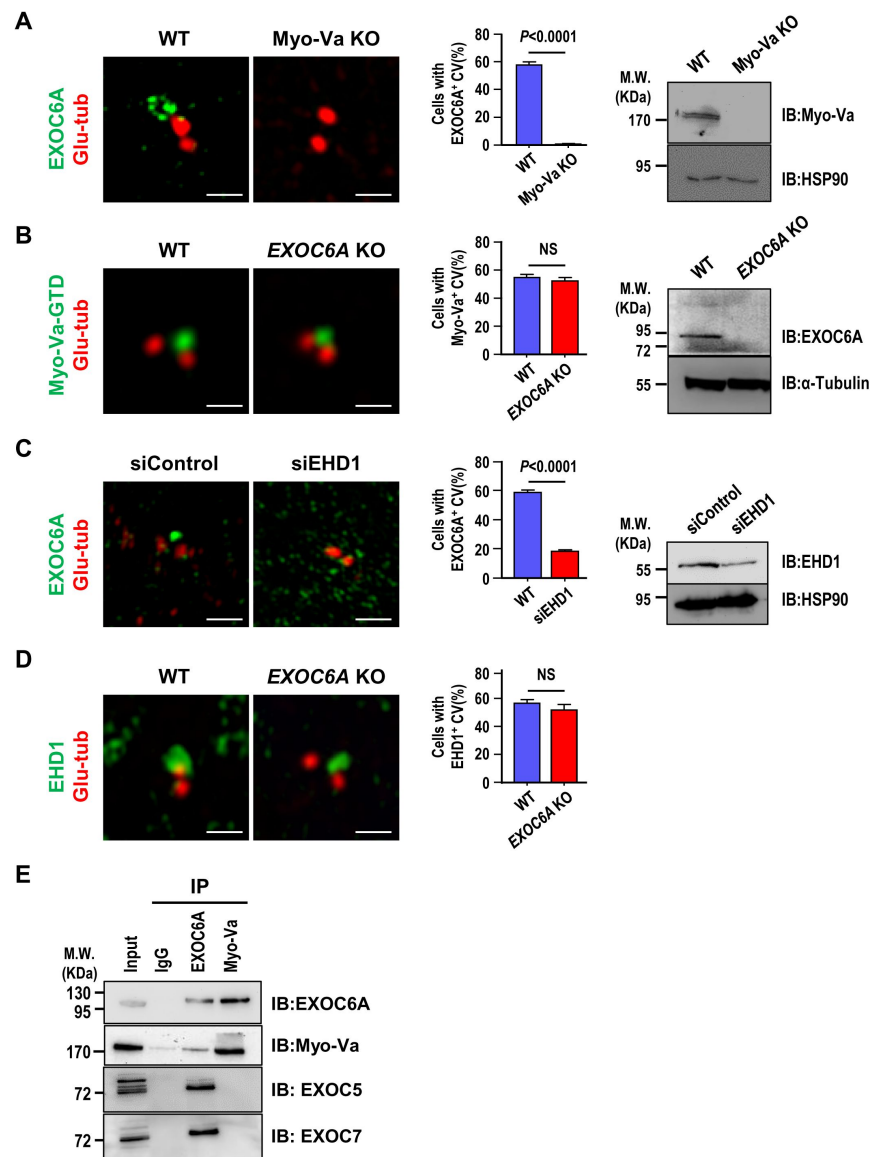


Figure 4.

Myo-Va and EHD1 are required for EXOC6A-labeled CV formation, and EXOC6A is associated with Myo-Va.

(A) Myo-Va is required for EXOC6A-labeled CV formation. RPE1-based *Myo-Va* KO cells and WT cells immunostained with antibodies against EXOC6A (green) and Glu-tub (red). Quantification of cells with EXOC6A-positive CVs (middle panel) and immunoblotting results (right panel) are shown. (B) EXOC6A is not required for Myo-Va-labeled CV formation. RPE1-based mCherry-Myo-Va-GTD-inducible cells against *EXOC6A* KO background and WT cells were immunostained with antibodies against Glu-tub (red). For easy differentiation, the mCherry-Myo-Va-GTD signal is converted to green. Quantification of cells with Myo-Va-associated CVs (middle panel) and immunoblotting results (right panel) are shown. (C) EHD1 is required for EXOC6A-labeled CV formation. RPE1 cells treated with siControl or siEHD1 for 48 h were immunostained with antibodies against EXOC6A (green) and Glu-tub (red). Quantification of cells with EXOC6A-positive CVs (middle panel) and immunoblotting results (right panel) are shown. (D) EXOC6A is not required for EHD1-labeled CV formation. RPE1-based *EXOC6A* KO and WT cells were immunostained with antibodies against EHD1 (green) and Glu-tub (red). Quantification of cells with EHD1-positive CVS is shown (right panel). (E) Co-IP experiments analyzed the association between EXOC6A and its potential binding proteins. Cell lysates were immunoprecipitated (IP) with EXOC6A or Myo-Va antibodies, followed by immunoblotting with the indicated antibodies. Error bars in A–D represent mean $\pm$ s.d. from at least 3 independent experiments with 100 cells per experiment. P-value was determined with two-tailed Student's t-test. $P < 0.05$ was considered statistically significant. NS, not significant. Scale bars are 1 μ m.

examined CP110 localization in *EXOC6A* KO cells. Our results showed that CP110 is appropriately removed from the distal end of the M-centriole in *EXOC6A* KO cells, indicating that *EXOC6A* is not required for this process (Fig. S3), which further support that *EXOC6A* is not essential for CV formation. Taken together, our findings suggest that Myo-Va transports a subset of *EXOC6A*-labeled vesicles to the existing CVs of the M-centriole for ciliary membrane assembly. However, *EXOC6A* is not essential for CV formation.

Transportation of *EXOC6A*-labeled vesicles to the M-centriole is via a dynein-, MT-, and Arp2/3-dependent pathway

Our previous study showed that the transportation of PCVs to the distal appendages of the M-centriole is mediated by Myo-Va, dynein, microtubules (MTs), and the Arp2/3 complex (Wu et al., 2018). In the current study, we found that Myo-Va and EHD1 are required for *EXOC6A*-labeled CV formation and that Myo-Va co-immunoprecipitates with *EXOC6A* (Fig. 4). It has been reported that dynein is necessary for the correct positioning of the Golgi complex near the centrosome upon serum starvation (Palmer et al., 2009; Wu et al., 2018), and that the inhibition of dynein activity with ciliobrevin D suppresses ciliogenesis (Firestone et al., 2012; Wu et al., 2018). Therefore, we investigated whether the transport of *EXOC6A*-associated vesicles to the DA of the M-centriole also requires dynein, MTs, and the Arp2/3 complex.

RPE1 cells were transfected with siRNAs targeting Golgin160, a protein required for the recruitment of dynein to the Golgi complex (Yadav et al., 2012), or treated with ciliobrevin D, a known cytoplasmic motor dynein inhibitor that suppresses ciliogenesis (Firestone et al., 2012), following the protocol described in Figure 5A. Our findings showed that depletion of Golgin160 led to a decrease in *EXOC6A*-labeled CV signals at or near the M-centriole (Fig. 5B, D) and resulted in the dispersion of Golgi matrix protein 130 (GM-130)-labeled Golgi complexes (Fig. 5B, E), ultimately leading to a reduction in the total number of cilia (Fig. 5F). Similar results were also observed in RPE1 cells treated with ciliobrevin D (Fig. 5C-i, D-F). We next treated RPE1 cells with nocodazole (NZ, an MT-depolymerizing drug) and examined its effect on ciliogenesis. Our results showed that the *EXOC6A*-labeled CVs and the cilia number are greatly reduced in NZ-treated cells (Fig. 5C-ii, D, and F). These data suggest that some *EXOC6A*-labeled vesicles originate from Golgi, and the transportation of *EXOC6A*-associated vesicles to the M-centriole requires dynein and microtubules.

Since Myo-Va-mediated transport of PCVs to the M-centrioles relies on an Arp2/3-dependent branched actin network surrounding M-centrioles (Wu et al., 2018), we next treated RPE1 cells with siARP2 (Fig. 5G). A reduced cilia number (Fig. 5G and H) and reduced *EXOC6A*-labeled CVs (Fig. 5G and I) were observed in siARP2-treated cells. A similar result was also found in cells treated with CK666 (a known specific Arp2/3 inhibitor) or high-dose (10 μ M) cytochalasin D (CytoD), which has been reported to reduce the association of ARP2 with the branched actin network (Fig. 5H-J). It has been reported that low concentrations (100 nM–1 μ M) of CytoD can promote ciliogenesis and increase cilia length (Cao et al., 2023; Kim et al., 2015; Kim et al., 2010; Wu et al., 2018), while high concentrations (10 μ M) lead to a significant inhibition of cilia formation (Wu et al., 2018). To further investigate the dose-dependent effects of actin cytoskeleton rearrangement on ciliogenesis, we treated RPE1 cells with CytoD at concentrations ranging from 200 nM to 10 μ M. Immunofluorescence staining of cilia using ARL13B (green) and Glu-tubulin (red) antibodies showed that low concentrations of CytoD (200 nM–2 μ M) resulted in significant cilia elongation, whereas higher doses (4 μ M, 7 μ M and 10 μ M) gradually inhibited cilia formation (Fig. S4). Consistent with our previous findings (Wu et al., 2018), our data showed that cilia length and cilia ratio increased at low doses of CytoD (200 nM–2 μ M) but gradually decreased with increasing doses (7 μ M and 10 μ M, Fig. S4B, S4C).

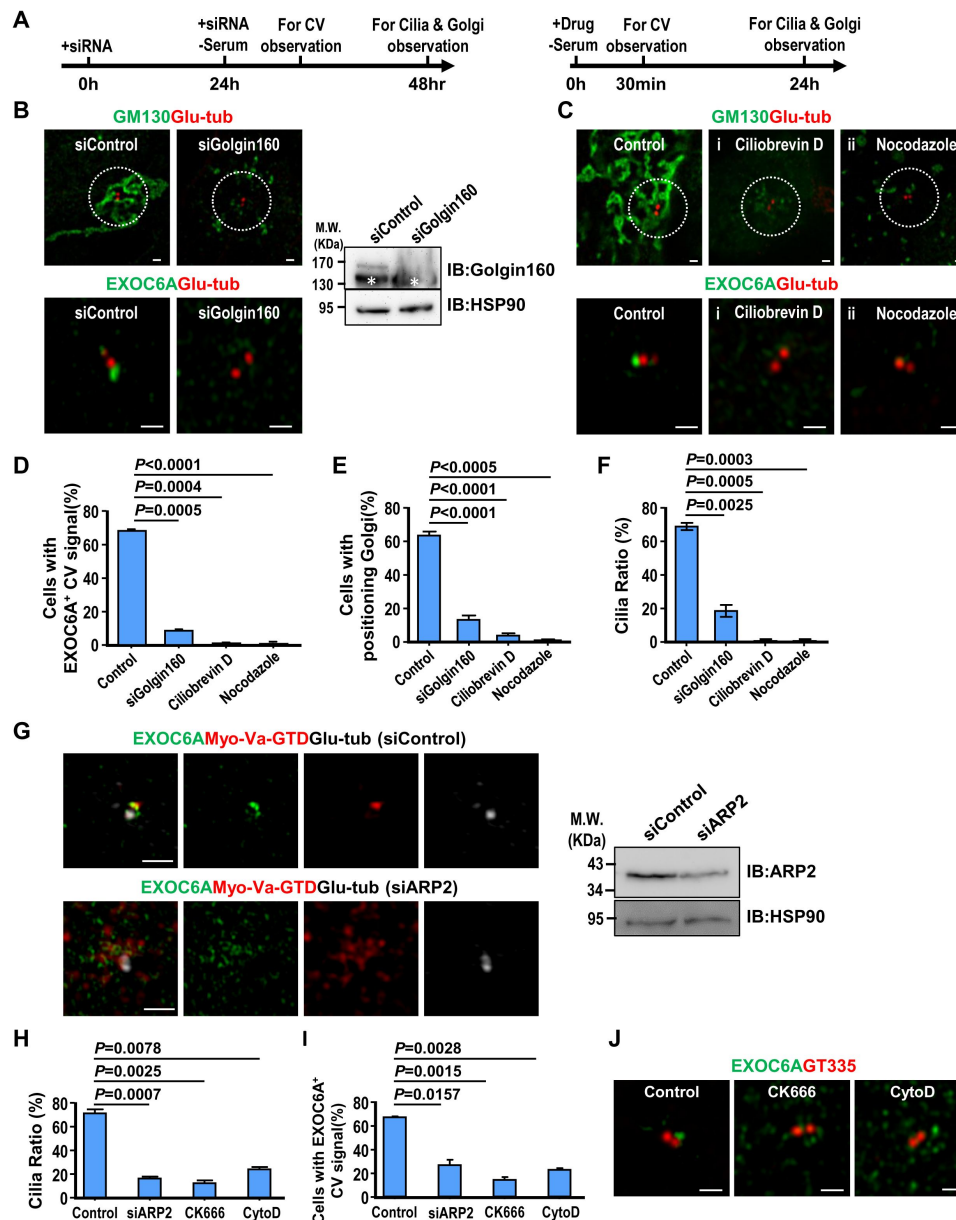


Figure 5.

Transportation of EXOC6A-associated vesicles to the M-centriole occurs via a dynein-, actin-, and MT-dependent pathway

(A) Schematic showing the protocols of siRNA (left) or drug (right) treatment. (B) RPE1 cells were treated with siControl or siGolgin160 and analyzed via confocal fluorescence microscopy or immunoblotting using the indicated antibodies. (C) RPE1 cells were treated with ciliobrevin D (i) or nocodazole (ii) and analyzed via confocal fluorescence microscopy using the indicated antibodies. (D-F) Cells with EXOC6A-associated CVs (D), a positioned Golgi (E), or a cilia ratio (F) were analyzed via 3D-SIM and quantified. Cells with a positioned Golgi were defined as those showing all GM130-labeled Golgi signals concentrated within a 7 μ m radius surrounding the Glu-tub-labeled centrioles. (G) RPE1-based mCherry-myosin-Va-GTD-inducible cells were treated with siARP2 as described in A (left) and analyzed via confocal microscopy or immunoblotting using the indicated antibodies. (H, I) Quantifications of the cilia ratio (H) and EXOC6A-associated CVs (I) is shown. (J) RPE1 cells were treated with CK666 or high-dose (10 μ M) CytoD as described in A (right panel) and analyzed via confocal microscopy using the indicated antibodies. Quantification of the cilia ratio (H) and EXOC6A-associated CVs (I) is shown. * in (B, right panel) indicates non-specific bands. Error bars represent mean $\pm$ s.d. from at least 3 independent experiments with 100 randomly selected cells. P-value was determined with two-tailed Student's t-test. $P < 0.05$ was considered statistically significant. NS, not significant. Scale bars are 1 μ m.

Taken together, our current studies reinforce our previous view (Wu et al., 2018) that low-dose of CytoD treatment (200 nM–2 μ M) can stabilize and/or enhance the formation of Arp2-associated branched actin filaments. However when cells were treated with high doses of CytoD (7 μ M and 10 μ M, Fig. S4) or the specific Arp2/3 inhibitor, CK-666 (Fig. 5), the association of Arp2 with the branched actin network was significantly reduced, thereby inhibiting ciliogenesis. In conclusion, our data suggest that Myo-Va carries EXOC6A vesicles to the DA of mother centriole via a dynein-, microtubule-, and actin-dependent pathway.

Loss of EXOC6A inhibits ciliogenesis, and some cells exhibit abnormal ciliary morphology when passing through the CV block

We found that EXOC6A signals are detected at CVs and the ciliary sheath (Fig. 1 and 2). We also found that EXOC6A-associated vesicles are dynamically recruited to or excreted from CVs and/or the ciliary membrane (Fig. 3). We next examined the effects of *EXOC6A* deletion on ciliogenesis. As shown in Figure 6, *EXOC6A* KO significantly inhibits cilia assembly, resulting in either no cilia or dotted CV signals after serum starvation (SS, 30 min–48 h). Upon serum starvation for 30 min, Myo-Va-GTD-labeled CVs were detected at the M-centriole in both wild-type control and *EXOC6A* KO cells (Fig. 6A). In control cells, the Myo-Va-GTD signal was mainly detected at the ciliary sheath membrane (Fig. 6A, SS 2h) 2 h after serum starvation, while the Myo-Va-GTD signals dissociated from the ciliary sheath membrane after longer starvation (SS 24 h or SS 48 h). Interestingly, most Myo-Va-GTD signals in *EXOC6A* KO cells were present as dotted CVs after longer starvation (SS 48 h), suggesting that ciliogenesis is arrested at the CV stage (Fig. 6A, lower panel).

We next used an antibody against ARL13B (a cilia shaft marker) to label the ciliary membrane and found that most *EXOC6A* KO cells exhibited no cilia (~45%, Fig. 6C) or dotted ARL13B signals (~50%) 2 h after SS (Fig. 6B; Fig. 6C, lower panel, SS 2 h), a pattern similar to Myo-Va staining (Fig. 6A, lower panel). Unexpectedly, after prolonged serum starvation, we could still detect approximately ~15% (SS 24 h) to ~30% (SS 48 h) of *EXOC6A* KO cells carrying ARL13B-labeled cilia protruding from the basal body. However, these ARL13B-labeled cilia showed a low intensity of ARL13B and Glu-tub signals (Fig. 6D and E) and the ARL13B-labeled cilia membrane exhibited a fragmented and punctuated staining pattern (enhanced in Fig. 6B, SS 48 h in lower panel). Our findings suggest that EXOC6A may play a crucial role in the further maturation of CVs into the ciliary sheath/shaft membrane during ciliogenesis.

To examine whether EXOC6A is required for ciliogenesis and whether its function can be rescued, we analyzed cilia formation in wild-type (WT), *EXOC6A* KO, and *EXOC6A* KO cells exogenously expressing GFP-EXOC6A. Immunofluorescence staining of cells with ARL13B and Glu-tubulin antibodies showed a significant reduction in the number of cilia in *EXOC6A* KO cells, while exogenous expression of GFP-EXOC6A in the KO cells can partially rescue cilia formation (Fig. S5).

Electron microscopy was further used to investigate the cilia morphology in the *EXOC6A* KO cells after prolonged serum starvation (72 h). To examine ciliary structure in *EXOC6A* KO cells in more detail, we used a 72 h-serum starvation protocol, which provided sufficient time for cilia assembly. Our results showed a perturbation in cilia assembly in *EXOC6A* KO cells (Fig. 6F and G). When compared with the wild-type control (Fig. 6F-i and ii), cilia in *EXOC6A* KO cells exhibited some ciliogenic defects, such that ciliogenesis was often arrested at the DA/PCV or CV stage (Fig. 6F-iii and iv), and some exhibited morphologically curved cilia bending in the TZ region (Fig. 6F-vi, 4/37 in Fig. 6G). Intriguingly, a small proportion of KO cells produced an elongated large CV in the absence of an extended axoneme (Fig. 6F-v, 4/37 in Fig. 6G). Consistent with this finding, curved cilia (Fig. 6B, SS 24 h, lower panel) and a low intensity of Glu-tub-labeled axoneme were observed in *EXOC6A* KO cells via confocal immunofluorescence microscopy (Fig. 6B, SS 48 h, lower panel, Fig. 6E). Nonetheless, we still observed 3 out of 37 morphologically normal-like cilia in *EXOC6A* KO cells (Fig. 6F-vii, 6G). Taken together, our data suggest that depletion of

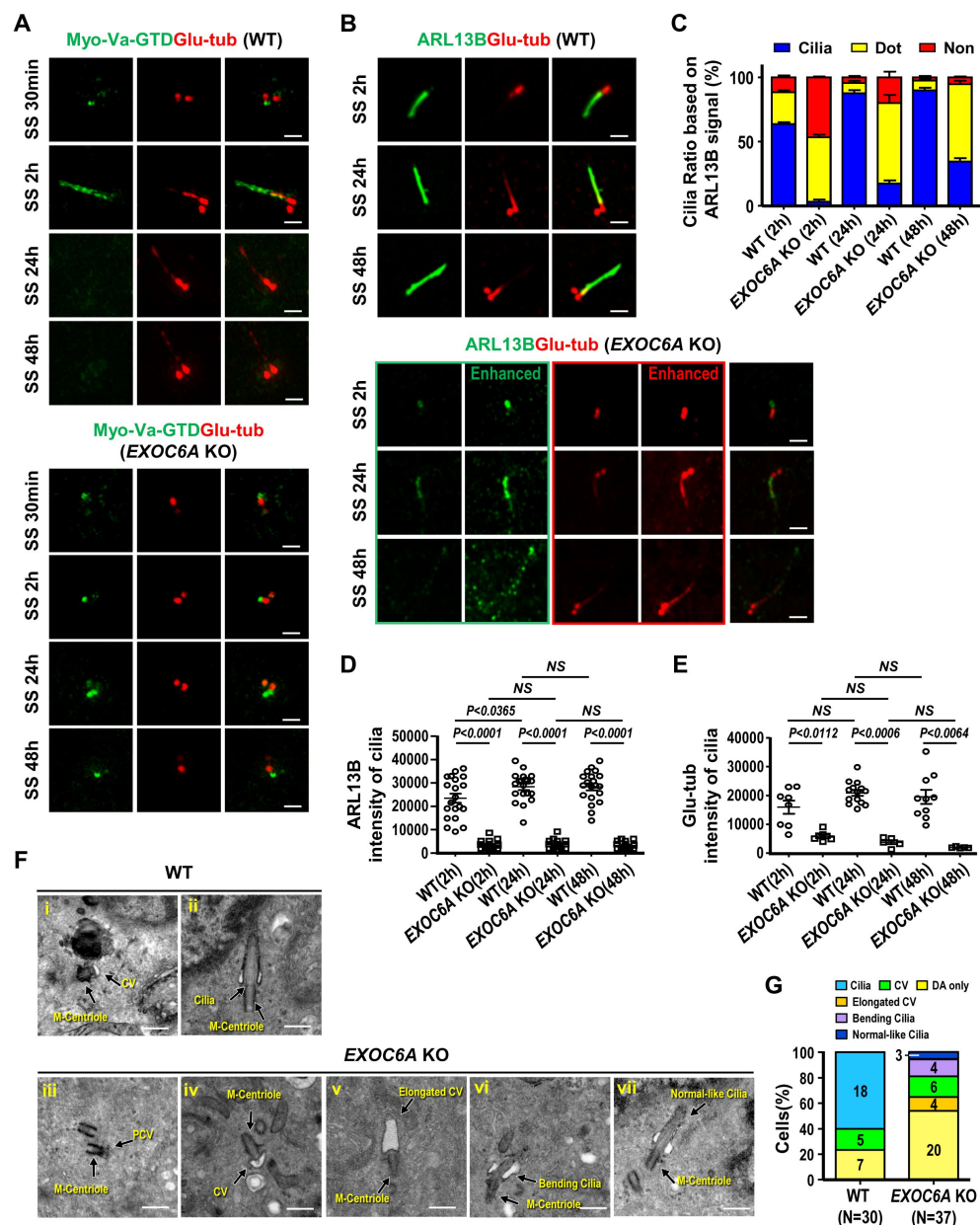


Figure 6.

Loss of EXOC6A inhibits ciliogenesis, and some cells exhibit an abnormal ciliary morphology when passing through the CV block.

(A-C) RPE1-based mCherry-Myo-Va-GTD-inducible WT and *EXOC6A* KO cells were treated with Dox for 24 h, serum starved (SS) at different time points, and analyzed via confocal fluorescence microscopy with the indicated antibodies. mCherry-Myo-Va-GTD signal is artificially converted to green for discrimination. (B) RPE1 WT and *EXOC6A* KO cells were serum starved at different time points and analyzed via confocal fluorescence microscopy with the indicated antibodies. For comparison, the intensities of ARL13B and Glu-tub of *EXOC6A* KO cells with non-enhanced and enhanced signals are shown in B (lower panel). (C) Cilia ratio measured through ARL13B labeling (non-enhanced signal). (D and E) Quantitation of non-enhanced ARL13B (D) and non-enhanced Glu-tub intensity on cilia (E) of WT or *EXOC6A* KO cells is shown. (F) WT and *EXOC6A* KO cells were serum starved for 72 h. Morphologies of normal or abnormal cilia, including DA, CVs, and ciliary membranes, were examined via EM (F) and quantified (G; N is the number of cells examined). Error bars in C, D, and E represent mean $\pm$ s.d. from at least 3 independent experiments from 100 randomly selected cells. P-value was determined with two-tailed Student's t-test. $P < 0.05$ was considered statistically significant. NS, not significant. Scale bars are 1 μ m in A and B. Scale bars are 500 nm in F.

EXOC6A primarily affects cells exhibiting no CV (showing only DA/PCV; [Fig. 6F-iii](#), [Fig. 6G](#)) and/or arrests cells at the CV stage, while once it passes through CV stage after prolonged serum starvation, it may interfere with subsequent cilia membrane formation at later stages of ciliogenesis.

EXOC6A is required to recruit NPHP and MKS module components, but not CEP290, to the TZ

Our EM study revealed both bending cilia at the TZ (4/37, [Fig. 6G-vi](#)) and morphologically normal-like cilia (3/37, [Fig. 6G-vii](#)). Intriguingly, immunofluorescence analysis revealed that the ARL13B signal intensity was greatly reduced in the normal-like cilia membrane in *EXOC6A* KO cells ([Fig. 6D](#)), suggesting a possible defect in the TZ of cilia. The TZ is a diffusion barrier at the ciliary membrane that acts as a gate to regulate the entry and exit of ciliary proteins required for signal transduction ([Jensen and Leroux, 2017](#)). Previous studies have identified three conserved protein modules (MKS, NPHP, and CEP290), composed of numerous protein complexes that cooperate in the assembly and gating function of the TZ ([Gonçalves and Pelletier, 2017](#)). CEP290, located at a proximal position near the M-centriole, acts as a hub that interacts with the NPHP and MKS module complexes ([Garcia-Gonzalo and Reiter, 2017](#)).

To verify the integrity of these abnormal cilia in *EXOC6A* KO cells, we examined the recruitment of several known TZ proteins, including CEP290, NPHP, and MKS module proteins. Our results showed that the recruitment of CEP290 to the ciliary base was normal in all examined ciliated cells, indicating that *EXOC6A* deletion did not affect CEP290 targeting to the TZ ([Fig. 7A](#)). However, the recruitment of several MKS complex proteins (TCTN2, AHI1, MKS3, and MKS1) to the basal body (M-centriole) was severely impaired in *EXOC6A* KO ciliated cells ([Fig. 7B](#)), while NPHP8 recruitment was only partially affected ([Fig. 7B](#)). Collectively, our findings suggest that *EXOC6A* is not required for CEP290 targeting to the TZ but plays an essential role in the recruitment of several NPHP and MKS module proteins to the basal body, which potentially affects the gating function of the TZ in *EXOC6A* KO cells.

Discussion

Cilia assembly is a complex, multistep process involving the accumulation and attachment of PCVs at the distal appendage, CV formation, and the subsequent extension of CVs into the ciliary sheath, alongside the formation of the ciliary shaft, ultimately leading to the mature cilia structure ([Breslow and Holland, 2019](#); [Long and Huang, 2019](#); [Nachury et al., 2010](#)). Currently, our understanding of the molecular mechanisms governing these steps remains incomplete.

Exocytosis is a process where intracellular secretory vesicles fuse with the plasma membrane, aiding in delivering integral membrane proteins to the cell surface and releasing materials into the extracellular space ([Martin-Urdiroz et al., 2016](#)). Human EXOC6A exhibits a high degree of similarity to the product of the *Saccharomyces cerevisiae* *SEC15* gene. In yeast, SEC15 is essential for the movement of vesicles from the Golgi apparatus to the cell surface ([Bowser and Novick, 1991](#)). Previous studies have indicated that SEC15 interacts with Myo2p (the yeast homolog of human Myo-Va) and plays a role in vesicle exocytosis ([Donovan and Bretscher, 2012](#); [Donovan and Bretscher, 2015](#)). Recently, exocyst was found to mediate the recycling of internal cilia and depletion of EXOC6A/Sec15a, but not other exocyst components, leads to a reduction in ciliogenesis ([Rivera-Molina et al., 2021](#)). This finding suggests that EXOC6A may have a unique function for ciliogenesis. However, the relationship and underlying mechanism between EXOC6A and Myo-Va in the context of ciliogenesis remain unclear.

In this study, we elucidated the role of the exocyst component EXOC6A in ciliogenesis and produced evidence for the model by which Myo-Va mediates trafficking of EXOC6A vesicles to the DA or existing CV of M-centriole in the early stages and to the ciliary membrane in the later stages

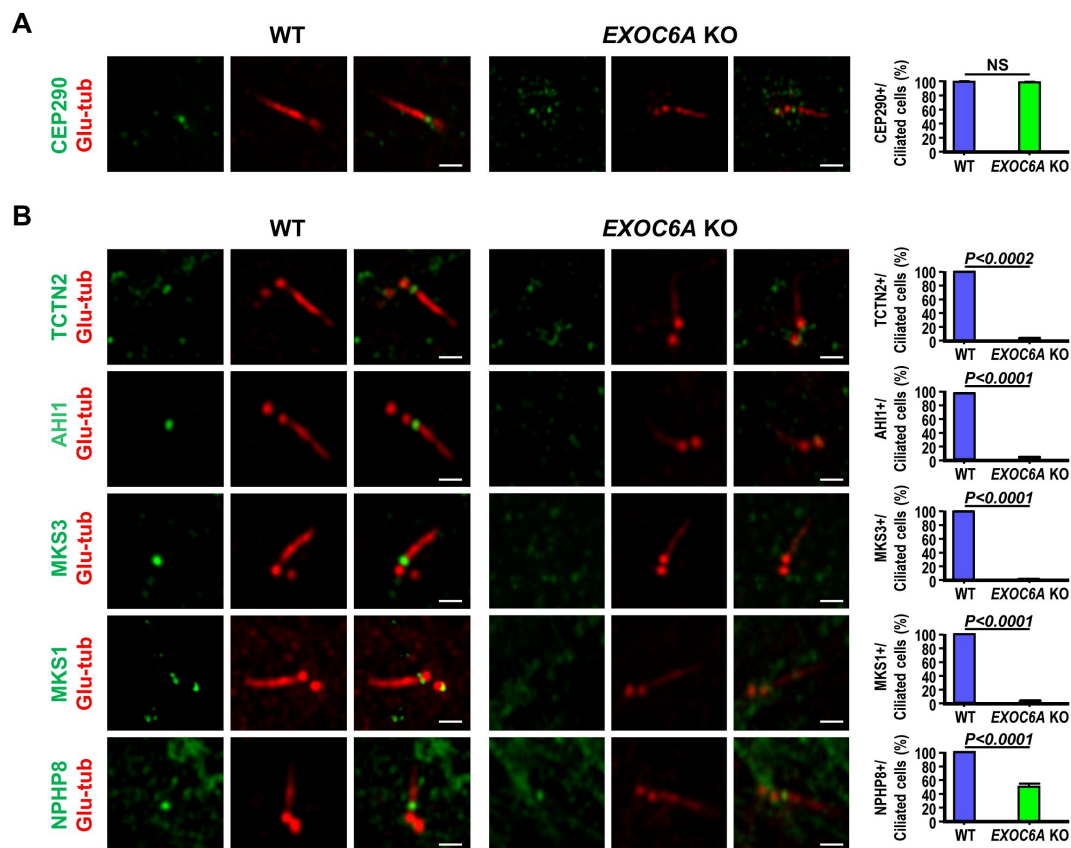


Figure 7.

EXOC6A is required for the recruitment of MKS module proteins to the peripheral TZ.

(A) WT and *EXOC6A* KO cells were fixed for 2 h (A) or 24 h (B) after serum starvation and analyzed via confocal fluorescence microscopy using antibodies against TZ proteins. Glu-tub staining labeled the centriole. Quantification of the fluorescence signals of various MKS proteins and NPHP8 on ciliated WT or *EXOC6A* KO cells is shown in the right panel. Error bars represent the mean $\pm$ s.d. from at least 3 independent experiments with 100 randomly selected cells per experiment. P-value was determined with two-tailed Student's t-test. $P < 0.05$ was considered statistically significant. NS, not significant. Scale bars, 1 μ m.

of ciliogenesis (**Figure 8**). We found that a subset of EXOC6A-positive signals colocalized with Myo-Va-labeled signals during the early phase of ciliogenesis, and that they exhibited a high correlation during clustering-to-centriole progression after initiating ciliogenesis (Fig. S1A-D). Intriguingly, not all EXOC6A-labeled signals were positive for Myo-Va (**Fig. 2A**), suggesting the presence of at least two populations of Myo-Va-labeled vesicles in cells: one that is EXOC6A+/Myo-Va+ and another that is EXOC6A-/Myo-Va+ (**Fig. 2A** and **Fig. 8**).

In this model, ciliogenesis begins shortly after serum-starvation-induced quiescence. Initially, Myo-Va mediates the transport of PCVs (EXOC6A-/Myo-Va+ and EXOC6A+/Myo-Va+ vesicles) to the DA of the mother centriole, followed by the formation of PCVs to a CV via EHD1 (Lu et al., 2015; Wu et al., 2018). However, EXOC6A is not absolutely required for CV formation, as EXOC6A KO did not interfere with the formation of Myo-Va- or EHD1-labelled CV (**Fig. 4B**, 4D). Once the CV is formed, the transport of EXOC6A vesicles to the existing CV in the early stages of ciliogenesis is still mediated by Myo-Va (**Fig. 4A**). The above process involves a two-step transportation mechanism: initially, dynein mediates the movement of Myo-Va-associated vesicles (either EXOC6A-positive or EXOC6A-negative) towards the pericentrosomal region along microtubules (MTs) (**Fig. 5B-F**). Subsequently, Myo-Va takes over and drives these vesicles (EXOC6A-positive or -negative) from the pericentrosomal region to the DAs of the M-centriole. This process is accomplished through the utilization of the ARP2/3-associated branched actin network (**Fig. 5G-J**). Notably, both of these Myo-Va-labeled vesicles require the function of EHD1 to form a CV (**Fig. 4D**; Wu et al., 2018). Our findings that EHD1 depletion (**Fig. 4D**) and Myo-Va KO (**Fig. 4A**) severely impaired the formation of EXOC6A-positive CVs at the M-centriole, suggest that Myo-Va and EHD1 play a role in promoting the formation of EXOC6-positive CVs during the early stages of ciliogenesis.

Although EXOC6A is not absolutely required for the initial CV formation (**Fig. 4B**, 4D), our data revealed another role of EXOC6A in the later stages of ciliogenesis. Our FRAP analysis and live cell imaging showed that a subset of EXOC6A-labeled vesicles displayed high dynamism throughout ciliogenesis, continually being recruited to, fusing into, or exited from existing CVs in the early stages and/or the ciliary membrane (or ciliary pocket) in the later stages of ciliogenesis (**Fig. 3**). These EXOC6A-positive vesicles, likely derived from the Golgi or recycling endocytic vesicles, are transported to the ciliary membrane (or ciliary pocket) and/or fused with/exited from the ciliary membrane in the later stages of ciliogenesis. The exocyst has been implicated in the transport of vesicles carrying ciliary proteins to the cilia (Wu and Guo, 2015). Thus, our findings suggested a potential role of EXOC6A vesicles, which may supply or remove proteins or lipid moieties from the CVs and/or the ciliary membrane (or ciliary pocket) during ciliogenesis.

Additionally, our results showed that EXOC6A KO led to the majority of the cells being arrested at the CV stage or exhibiting no or few cilia (**Fig. 6C**); only a small proportion of cilia were able to elongate, exhibiting an abnormal ciliary morphology (**Fig. 6B**, 6F). Electron microscopic analysis of these abnormal cilia revealed elongated CV (lacked axoneme, **Fig. 6F-v**), curved cilia bending at the TZ region (**Fig. 6F-vi**), or normal-like cilia (**Figure 6B-vii**), whose structure and membrane composition need to be further characterized). Further confocal immunofluorescence analysis revealed a curved ciliary morphology (**Fig. 6B**, SS24h) and impaired recruitment of several known TZ proteins, including TCTN2, AHI1, MKS3, and MKS1, to the basal body in EXOC6A KO cells (**Fig. 7**). Taken together, our findings suggest that EXOC6A is required for the early and the later stages of ciliogenesis as providing or removing ciliary components (proteins or membrane moieties) to the existing CV or ciliary membrane, which highlights its multifaceted involvement in the process of ciliary membrane formation.

Furthermore, the Rab11-Rabin8-Rab8 signaling pathway, which involves the recruitment of Rabin8 (a GDP-GTP exchange factor for Rab8) and subsequent activation of Rab8, has been implicated in ciliary membrane assembly (Knödler et al., 2010; Nachury et al., 2007; Westlake et al., 2011; Yoshimura et al., 2007). Recently, RAB34, located at the ciliary sheath (Ganga et al.,

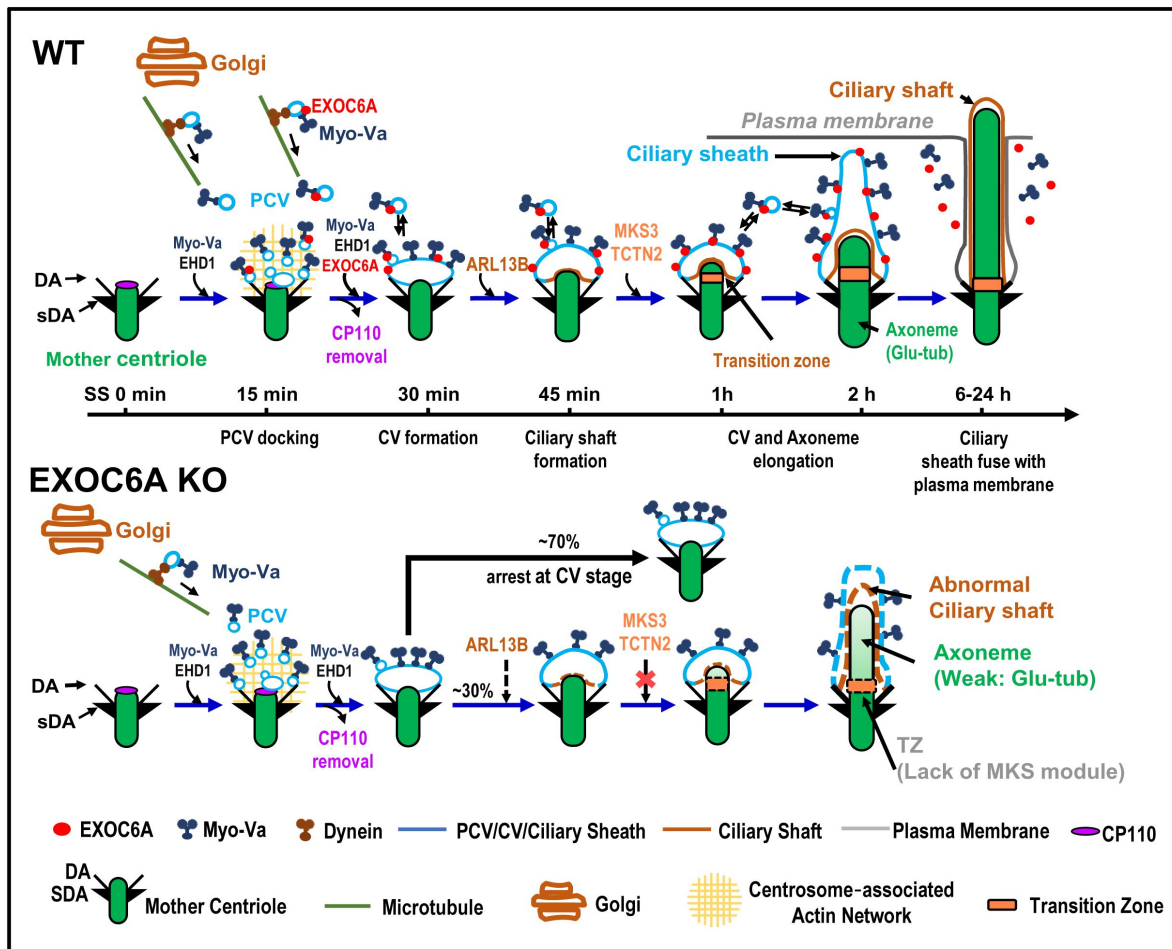


Figure 8.

Ciliogenesis model and comparison between WT and *EXOC6A* KO cells.

EXOC6A is located at the PCV/CV/ciliary sheath, and Myo-Va mediated the transportation of *EXOC6A* vesicles to the mother centriole via a dynein-, microtubule-, and actin-dependent pathway. *EXOC6A*-associated vesicles are continuously recruited and fused or excreted from the CVs or ciliary membrane. Depletion of *EXOC6A* impairs ciliogenesis, and most cells are arrested at the CV stage. There is a lack of MKS module proteins of the TZ in cilia with defects in *EXOC6A* KO cells. PCV, preciliary vesicle; CV, ciliary vesicle; DA, distal appendage; sDA, subdistal appendage.

2021 [Stuck et al., 2021](#)), was also reported to be required for ciliogenesis (Xu et al., 2018). The relationship between EXOC6A, Rab11 signaling, and RAB34 in ciliogenesis is not clear. Future experiments dissecting the underlying mechanisms between Rab11 signaling, RAB34, and EXOC6A could provide a more complete picture of ciliary membrane assembly during ciliogenesis.

Finally, ciliopathy-associated genes have been demonstrated to impact various stages of mouse cortical development (Guo et al., 2015), and severe human ciliopathies are linked to developmental abnormalities in the forebrain (Andreu-Cervera et al., 2021). Mutations or deletions in genes encoding exocyst proteins, including *EXOC2*, *EXOC4*, *EXOC7*, and *EXOC8*, have been associated with ciliopathies and neural developmental disorders (Coulter et al., 2020; Dixon-Salazar et al., 2012; Shaheen et al., 2013; Turkyilmaz et al., 2021; Van Bergen et al., 2020). Interestingly, deletions involving the 5' portion of the *EXOC6A* gene and two adjacent cytochrome p450 genes (*CYP26A1* and *CYP26C1*) are associated with autosomal-dominant nonsyndromic optic aplasia (ONA), an extremely rare disorder that causes unilateral or bilateral blindness in the affected eye (Meire et al., 2011). The molecular basis remains unclear. Further studies are needed to explore the specific roles of various types of ciliary vesicles, the underlying mechanisms of their trafficking/fusion, and their relevance to ciliopathies and neural developmental disorders.

In summary, we have identified a novel role for EXOC6A in ciliogenesis and determined that it is essential for both early and late stages of this process. EXOC6A vesicles may transport proteins or lipid moieties to and from the CV (early phase) and the ciliary pocket/ciliary membrane (late phase) during ciliogenesis. However, EXOC6A is not absolutely necessary for the initial CV formation; its function may be to help assemble the complete transition zone and promote the maturation of existing CVs into the ciliary sheath/shaft membrane. Furthermore, transportation of EXOC6A vesicles to the mother centriole is mediated by Myo-Va and occurs via a dynein-, MT-, and Arp2/3-dependent pathway. Our study provides new insights into the function of EXOC6A in cilia membrane assembly during ciliogenesis. Future studies are needed to explore the specific roles of different types of ciliary vesicles, the underlying mechanisms of their trafficking and fusion, and their relevance to ciliopathies and neural developmental disorders.

Methods

Plasmids and antibodies

cDNAs encoding full-length EXOC6A were obtained via RT-PCR using total RNAs from human HEK293T cells and subcloned in-frame into pcDNA4/TO/myc-His-A (Invitrogen) or pLVX-Tight-Puro (BD Biosciences Clontech) GFP expression vectors. GFP- and mCherry-myosin-Va-GTD constructs have been previously described (Wu et al., 2018). The sequences of all constructed plasmids were confirmed.

Commercial antibodies used were anti-Sec15 (clone [15S2G6], 1/100 for IF, Kerafast), anti-EXOC6 (1/100 for IF and 1/2000 for WB, Novus), anti-EXOC5 (1/2000 for WB, Proteintech), anti-EXOC7 (1/2000 for WB, Proteintech), anti-polyglutamylated tubulin (Glu-tub, GT335, 1/100 for IF, AdipoGen), anti-CEP164 (1/500 for IF, Novus), anti-GFP (Abcam and Clontech), anti-ARL13B (1/600 for IF, Proteintech), anti-myosin-Va (1/500 for IF and 1/5000 for WB, Novus), anti-EHD1 (1/500 for IF and 1/5000 for WB, Novus), anti-GM130 (1/1000 for IF, Abcam), anti-CP110 (1/600 for IF, Proteintech), anti-Golgin160 (1/2000 for WB, Novus), anti-Arp2 (1/1000 for WB, Abcam), anti-TMEM67 (MKS3, 1/400 for IF, Proteintech), anti-TCTN2 (1/400 for IF, Proteintech), anti-AHI1 (1/400 for IF, Proteintech), and anti-MKS1 (1/400 for IF, Proteintech). The secondary antibodies were Alexa 488/568/647 conjugated anti-mouse or rabbit (Invitrogen). The rabbit polyclonal antibody against CEP120 (residues 639–986) was as described in our previous paper (Lin et al., 2013; Tsai et al., 2019).

Cell culture and cell lines

Human telomerase-immortalized retinal pigment epithelial cells (RPE1) (Wu et al., 2018 [DOI](#)) originally obtained from ATCC were maintained in DMEM/F12 (1:1) supplemented with 10% FBS. Cells were transfected with various cDNA constructs using TransIT-LT1 Transfection Reagent (Mirus). All cell lines were tested for mycoplasma contamination and found negative.

The gRNA expression plasmids were obtained from Addgene (Addgene plasmid # 48138; <http://n2t.net/addgene:48138> [DOI](#); RRID:Addgene_48138), having been generated by inserting annealed primers into the gRNA cloning vector pSpCas9-2A-GFP (PX458) (Ran et al., 2013 [DOI](#)). The targeting sequences for the EXOC6A gRNA (gRNA1:5'-AAGTTACTGATACCAACCGA-3' and gRNA2:5'-TTGTATCCGTAATCATGACA-3') primer pair purchased from Mission Biotech were annealed and cloned into PX458. For transfection, 2.5 µg gRNA (PX458 plasmid) was transfected with LT-1 reagent (Mirus) into RPE1 cells, according to the manufacturer's instructions. The transfected cells were sorted with GFP signals using a cell sorter (FACSARIA, BD Biosciences) into single cells and expanded.

To confirm the depletion of EXOC6A, genomic DNAs isolated from *EXOC6A* knockout colonies were subjected to PCR using the following primers: 5'-TTGGGTCAGTGATTGAATTG-3' and 5'-CCAAATAATCTGTAATCCCATA-3' for gRNA1 and 5'-ACATCTCCTGAGCCTCATACC-3' and 5'-GCTTCAGAAAAAGAGAATACTCCT-3' for gRNA2. The sequences of PCR products were then confirmed. Also, the protein expression of expanded colonies was confirmed via IF and Western blotting.

RPE1-based doxycycline (DOX)-inducible cell lines were as described previously (Wu et al., 2018 [DOI](#)). To obtain RPE1 cell lines that inducibly expressed GFP-EXOC6A, GFP-Myo-Va-GTD, and mCherry-Myo-Va-GTD, lentiviruses generated using the target cDNAs in the pLVX-tight-puro vector were used to infect RPE1 Tet-On cells (stably expressing rtTA, Clontech). The infected cells were sterile-sorted using a cell sorter (FACSARIA, BD Biosciences) for GFP or mCherry, and the positive cells were selected and expanded as inducible cell lines.

The siRNAs and non-targeting siRNA control were transfected into cells using Lipofectamine RNAiMAX (Invitrogen), according to the manufacturer's protocol. To increase knockdown efficiency, two rounds of silencing were used before the experiments. The siRNA sequences for siEHD1 (5'-GGAGAGAUCUACCAGAAGA-3'), siGolgin-160 (5'-CAGCCUCCUUGCCGCGAGGGCCUC-3'), and siARP2 (5'-CCAGCUUUGGUUGGAAGACCUAUA-3') were purchased from Invitrogen.

Immunofluorescence confocal microscopy

Immunofluorescence confocal microscopy was carried out as previously described (Wu et al., 2018 [DOI](#)). Briefly, cells were grown on coverslips and fixed in methanol at -20 °C for 20 min. The cells were then blocked with 10% normal goat serum in PBST. After blocking, the cells were incubated with primary antibodies, washed with PBST, and then incubated with Alexa Fluor 488-, 568-, or 647-conjugated secondary antibodies (1/500; Invitrogen). The samples were visualized using an LSM880 Airyscan system (Carl Zeiss).

For the fluorescence recovery after photobleaching (FRAP) experiment, live imaging of RPE1-based GFP-EXOC6A and mCherry-myosin-Va-GTD cells was performed using the LSM880. A 405 nm laser was used to photobleach associated CVs and the ciliary sheath 100 times. The changes in fluorescence intensity of GFP-EXOC6A were measured from the projected z-stacks using ZEN software.

Super-resolution microscopy (3D-SIM)

The cells were fixed and stained as described (see above), and multi-colored beads (100 nm; Invitrogen) were added before mounting. Sixteen-bit super-resolution images with five phases and five rotations were obtained using a Zeiss ELYRA PS.1 LSM780 system or a Zeiss ELYRA7 system (Carl Zeiss). We used a plan-apochromatic 63×/1.4 NA oil-immersion M27 objective in combination with the default grid (stripe sim in ELYRA PS.1 and Lattice sim plus SIM square). We used the “optimized” setting for the z-stack interval. The raw images were reconstructed using ZEN Black (Carl Zeiss) under the default parameters. The reconstructed images were corrected with signals of beads for chromatic aberration in the x, y, and z directions, and the images were aligned accordingly with ZEN Black. We followed the standard guidelines (Demmerle et al., 2017 [DOI](#)) and used the SIM-Check software (Ball et al., 2015 [DOI](#)) to verify the raw, reconstructed, and calibrated data. To visualize centriole and axoneme structures during live-cell imaging, we employed SiR-tubulin, a fluorogenic probe that binds to endogenous microtubules (spirochrome).

Ultrastructure expansion microscopy (U-ExM)

U-ExM was used to reveal the fine cellular architecture of the cells. Cells were processed for this U-ExM as previously described (Gambarotto et al., 2019 [DOI](#)). In brief, cells were fixed with methanol or 4% PFA based on our target. First, cells on coverslips were incubated in a PBS solution containing 0.7% formaldehyde (FA; 36.5–38%; F8775; Sigma-Aldrich) plus 0.15% acrylamide (AA; 40%; A4058; Sigma-Aldrich) for 5 h at 37 °C. Next, the cells on coverslips were incubated in U-ExM monomer solution, composed of 19% (wt/wt) sodium acrylate (SA; 97–99%; 408220; Sigma-Aldrich), 10% (wt/wt) AA, and 0.1% (wt/wt) *N,N'*-methylenebisacrylamide (BIS; 2%; M1533; Sigma-Aldrich) at 37 °C for 1 h in the dark for gelation. We transferred coverslips with gel to 6-well plates, added 1 mL denaturing buffer (200 mM SDS, 200 mM NaCl, and 50 mM Tris in ultrapure water, pH 9), and shook for 15 min at room temperature (RT). We then separated gel from coverslips and transferred the gel to 1.5 mL centrifuge tubes with fresh denaturing buffer, then incubated at 95 °C for 30 min. After denaturation, the gels were placed in beakers filled with ddH₂O overnight for expansion. For antibody staining, the gels were PBS washed and then incubated with primary antibodies for 1 or 2 nights at RT (depending on the antibodies). The gels were washed in PBST (PBS plus 0.1% tween20) and incubated with a secondary antibody for 3 h at 37 °C. The gels expanded by around 4x after this treatment. The samples were then visualized using an LSM880 Airyscan system (Carl Zeiss).

TEM and CFLEM

Cells were processed for transmission electron microscopy and CFLEM as previously described (Reddick and Alto, 2012 [DOI](#)). Briefly, RPE1 and RPE1-based *EXOC6A* KO cells were grown on Aclar film (Electron Microscopy Sciences) and fixed in 2% glutaraldehyde (GA) containing 1% tannic acid in 0.1 M cacodylate buffer. The fixed cells were post-fixed with 1% OsO₄ in 0.1 M cacodylate buffer at room temperature (RT) for 30 min. Then, the cells were further stained with 1% uranyl acetate at RT for 1 h, dehydrated in a graded series of ethanol, infiltrated, and embedded in Spurr's resin. Ultrathin sections (70 nm) were stained with 4% uranyl acetate (UA) and Reynold's lead citrate for 10 min and examined with an electron microscope (Tecnai G2 Spirit TWIN, FEI).

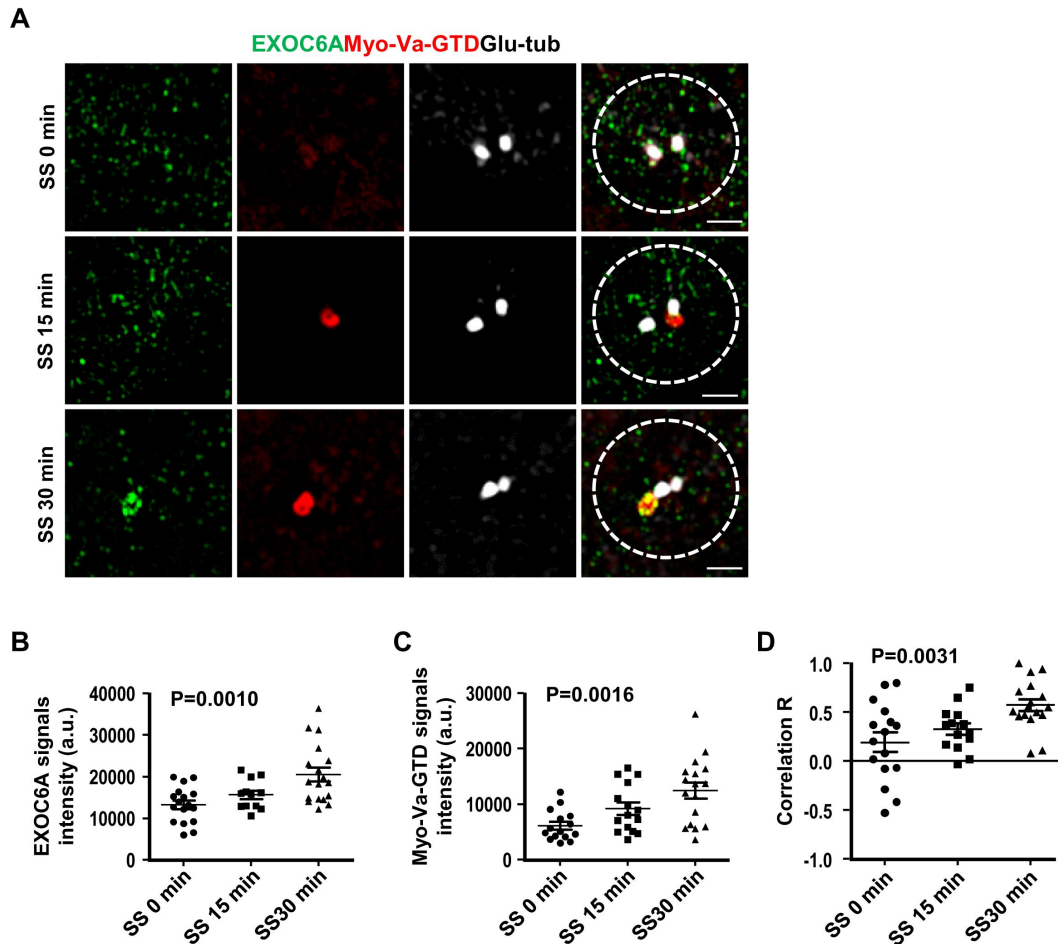
For CFLEM, RPE1-based GFP-*EXOC6A*-inducible cells were grown on laser-etched glass gridded coverslips affixed to the bottom of live-cell dishes (MatTek). Cells were treated with DOX for 24 h and serum starved for 2 or 24 h. The cells were then fixed with 2 % glutaraldehyde plus 1% tannic acid in 0.1 M cacodylate buffer, and the GFP-*EXOC6A* signals were imaged via 3D-SIM using a Zeiss ELYRA system (Carl Zeiss). The cells were then embedded in EPON resin. After EPON polymerization, resin blocks were detached from the glass dishes. Using the grid patterns imprinted in the resin, serial thin sections (70 nm) of the squares of interest were cut. The sections

were stained with 4% uranyl acetate and Reynold's lead citrate for 10 min, and the cells were examined with an electron microscope (Tecnai G2 Spirit TWIN, FEI). All EM images were processed using Gatan Digital Micrograph software (Gatan).

Statistics and reproducibility

Statistical analyses were performed using GraphPad Prism. Results are presented as mean $\pm$ standard derivation (s.d.), as specified in the figure legends. Sample size and the number of repeated experiments are also described in the legends. Statistical differences between two datasets were analyzed using the two-tailed paired Student's *t*-test. The precise *P*-values are shown in the figures. $P < 0.05$ was considered statistically significant. All experiments were repeated at least three times.

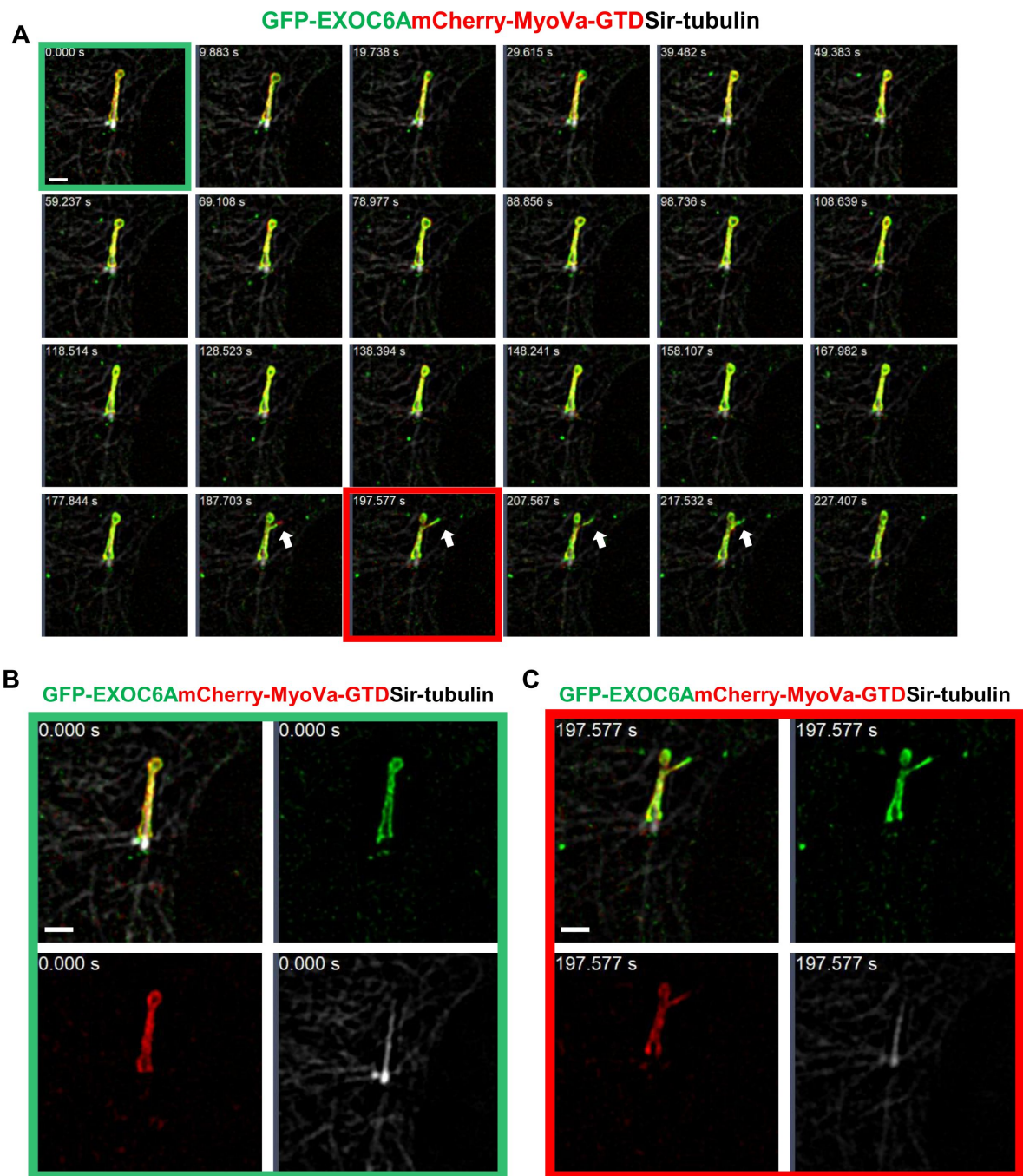
Supplementary figures



Supplementary Figure 1.

Correlation of spatial localization of EXOC6A and Myo-Va during the early stages of ciliogenesis.

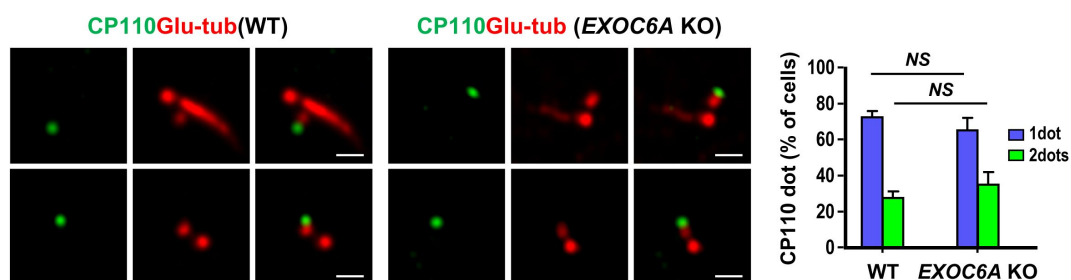
(A) RPE1-based mCherry-Myo-Va-GTD-inducible cells were treated with DOX for 24 h and serum starved for 0, 15, and 30 min. Fluorescence intensities of EXOC6A (green) or Myo-Va-GTD (red) signals within a 2 μ m radius surrounding the Glu-tub-labeled centrioles (white) were quantified and are shown in B (EXOC6A) and C (Myo-Va-GTD). Correlation R (Pearson correlation coefficient) between EXOC6A and Myo-Va is shown in D. Correlation R was analyzed using Zeiss Zen blue software. Error bars in B–D represent mean $\pm$ s.d. from at least 3 independent experiments with 100 randomly selected cells. P-value was determined with one-way ANOVA. $P < 0.05$ was considered statistically significant. NS, not significant. Scale bars are 1 μ m.



Supplementary Figure 2.

Live-cell imaging of GFP-EXOC6A co-localized with Myo-Va-GTD at the ciliary membrane

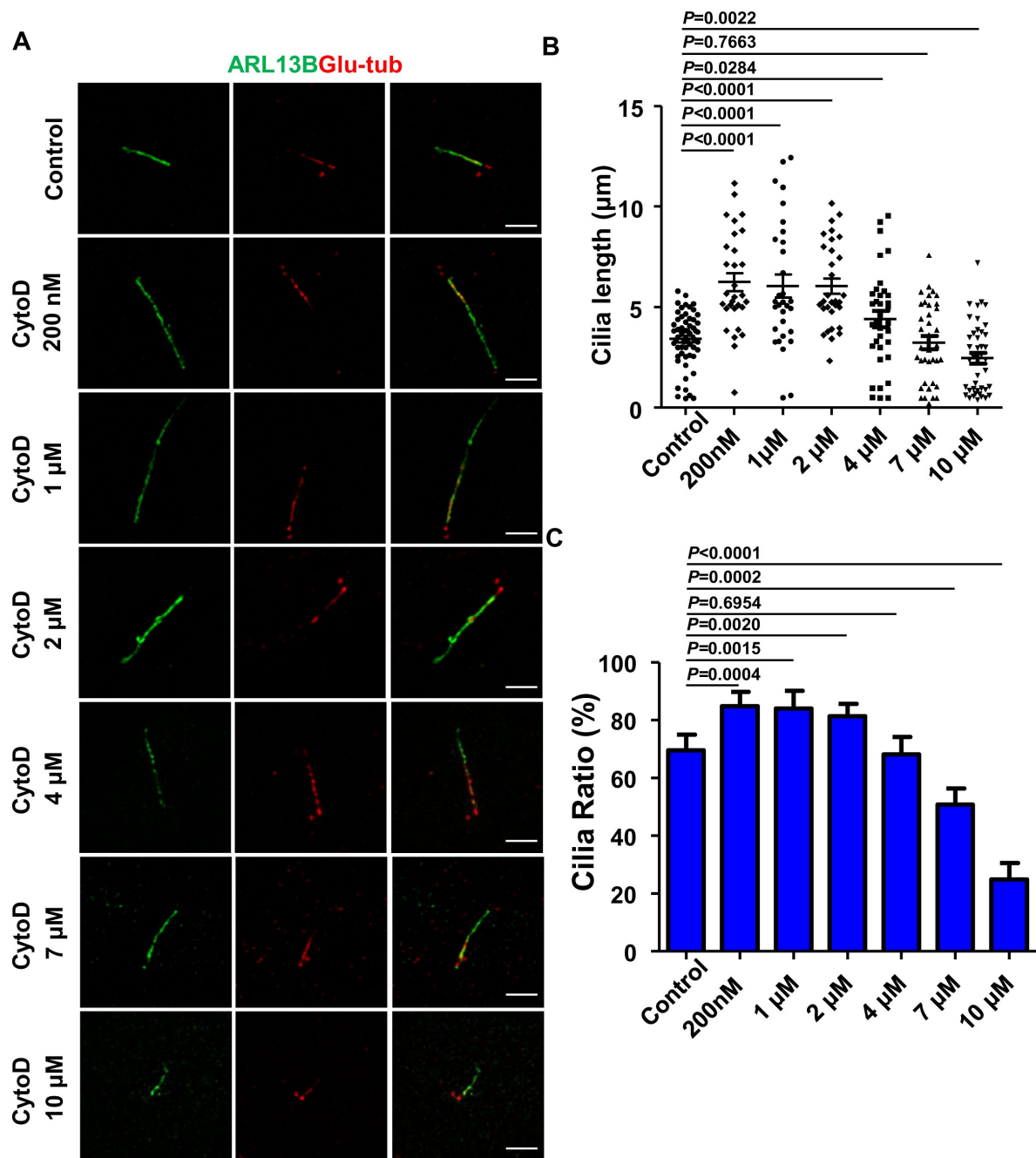
(A) RPE1-based inducible cells expressing GFP-EXOC6A and mCherry-MyoVa-GTD were treated with Dox for 24 h, serum starved for 30 min, and Sir-tubulin was added to label centriole and axoneme. (B) Single channel image (green box in A) captured at the starting time point (0 sec). (C) Single channel image (red box in A) captured at frame 21 (197.577 sec). Scale bars are 1 μ m.

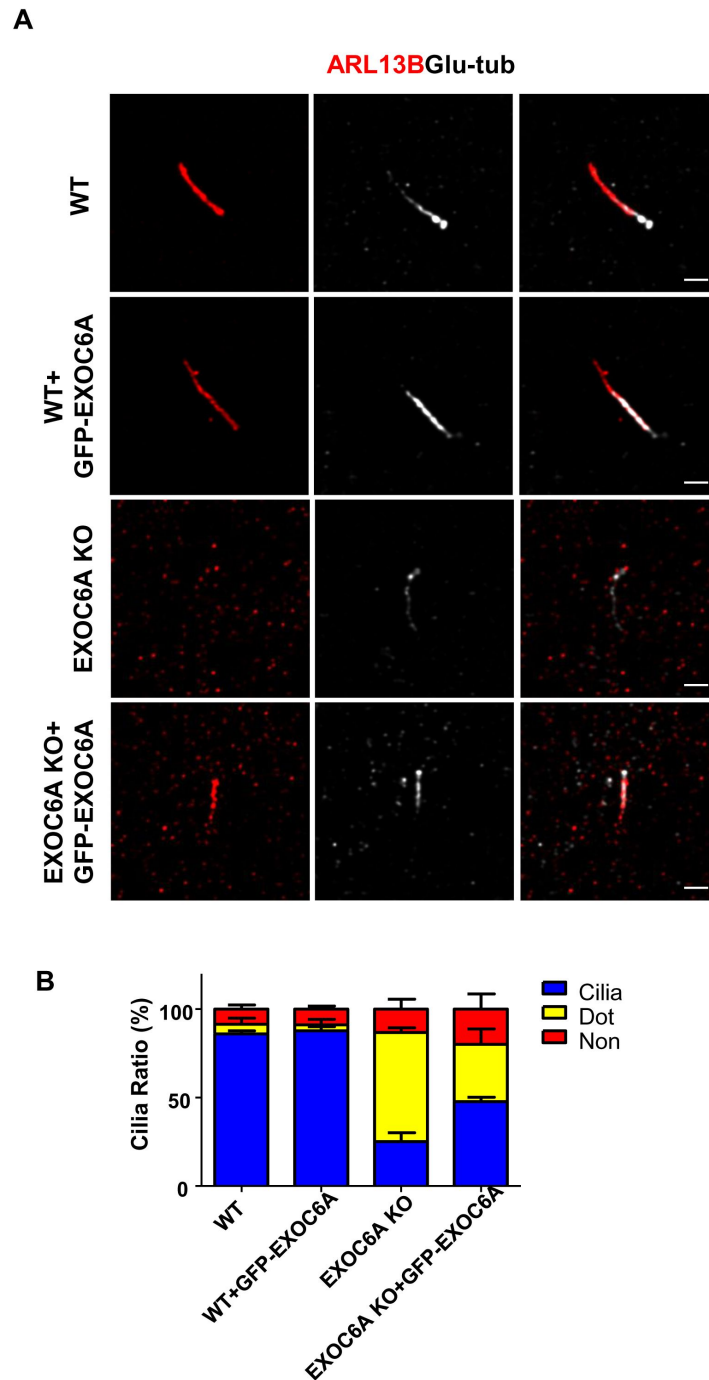


Supplementary Figure 3.

***EXOC6A* deletion does not interfere with the removal of CP110 from the mother centriole.**

WT and *EXOC6A* KO cells were fixed 24 h after serum starvation and analyzed via fluorescence confocal microscopy using the indicated antibodies. (B) Percentages of cells with one or two CP110 dots are shown. Error bars represent the mean $\pm$ s.d. from at least 3 independent experiments with 100 cells per experiment. P-value was determined with two-tailed Student's t-test. NS, not significant. Scale bars are 1 μ m.





Supplementary Figure 5.

Rescue of ciliogenesis defects in EXOC6A knockout cells by GFP-EXOC6A re-expression.

(A) Representative immunofluorescence images of RPE1 cells (WT, WT+GFP-EXOC6A, EXOC6A KO, and EXOC6A KO+GFP-EXOC6A) stained for ARL13B (red) and Glu-tubulin (white). **(B)** Quantitative analysis of ARL13B-labeled cilia phenotypes. Data are presented as the mean $\pm$ s.d. of 100 randomly selected cells from at least three independent experiments. Scale bars are 1 μ m.

Source Data

Uncropped images of Western blots shown in the main figures.

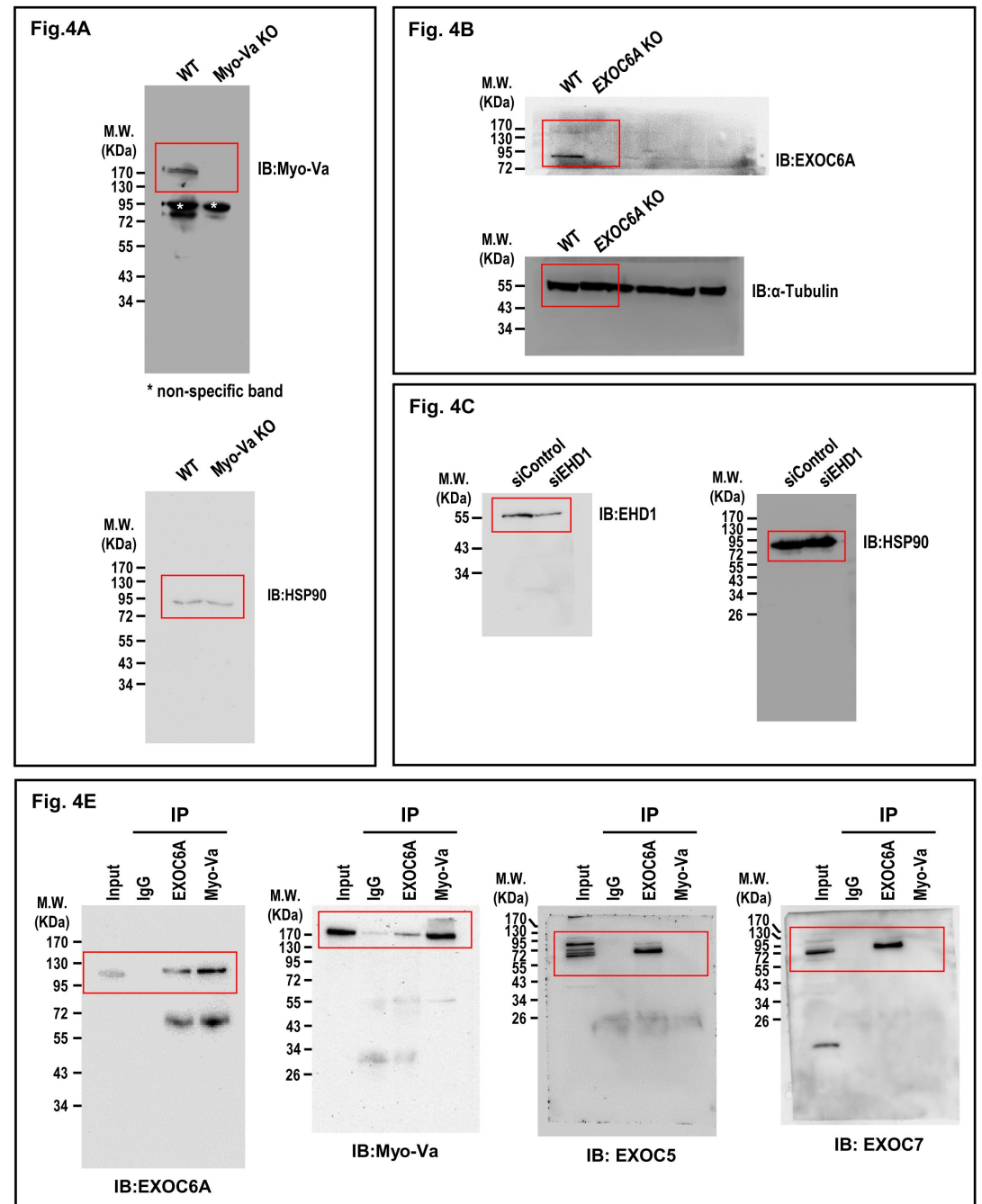


Fig. 5B

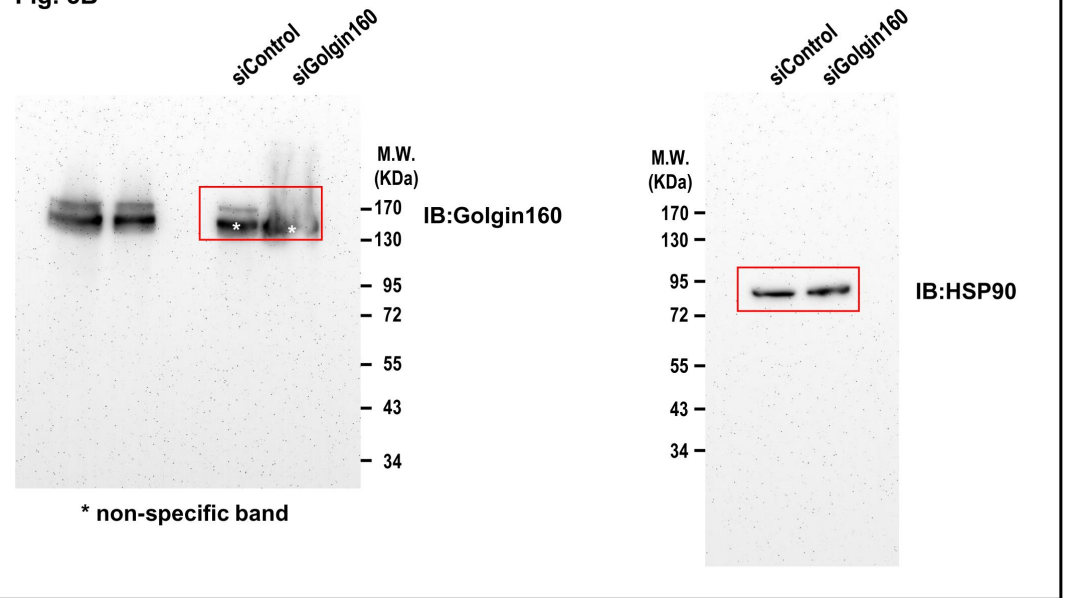
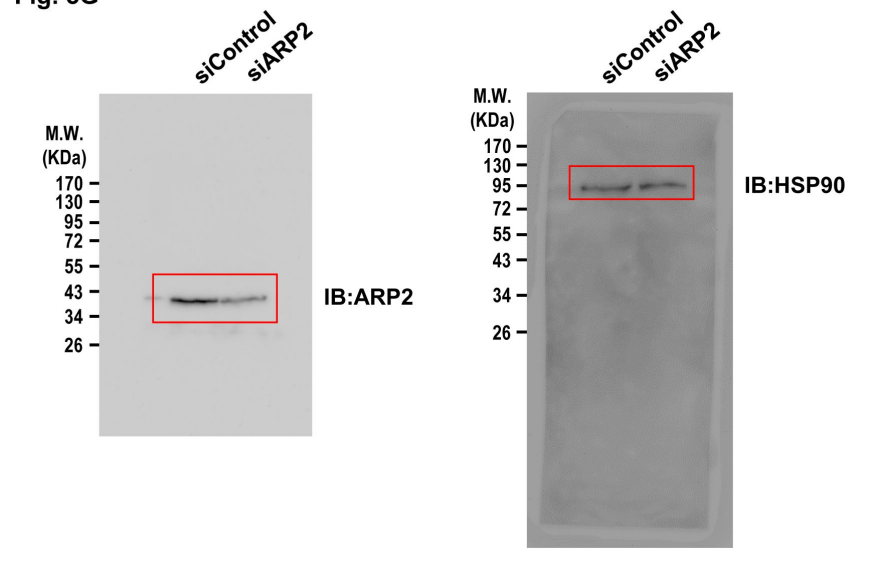


Fig. 5G



Acknowledgements

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Additional information

Author contributions

Conceptualization: TLL and TKT; Investigation: TLL and CTW; Methodology: TLL, CTW, and TKT; Resource: TLL and TKT; Original Draft: TLL and TKT; Writing: TLL and TKT; Review & Editing: TLL, CTW, and TKT.

Additional files

Video 1. RPE1-based inducible cells expressing GFP-EXOC6A and mCherry-Myo-Va-GTD were treated with Dox for 24 h and then serum starved for 30 min. Live cell images were taken using an LSM880 Carl Zeiss confocal system under a controlled CO₂ (5%) and temperature (37 °C). Photobleaching was performed on the CVs. The first time point of photobleaching the CVs was at 2 min, following which the CVs gradually recovered. Related to Fig. 3A. [🔗](#)

Video 2. RPE1-based inducible cells expressing GFP-EXOC6A and mCherry-Myo-Va-GTD were treated with Dox for 24 h and then serum starved for 30 min. Live cell images were taken using an LSM880 Carl Zeiss confocal system under a controlled CO₂ (5%) and temperature (37 °C). Photobleaching was performed on the ciliary membrane. The first photobleaching time point for the ciliary membrane was at 2 min, after which the ciliary membrane gradually recovered. Related to Fig. 3B. [🔗](#)

Video 3. RPE1-based inducible cells expressing GFP-EXOC6A and mCherry-MyoVa-GTD were treated with Dox for 24 h, then serum starved for 30 min, and SiR-tubulin was added to label centriole and axoneme. Live cell images were acquired using an Elyra 7 SIM Carl Zeiss confocal system under a controlled CO₂ (5%) and temperature (37 °C). Related to Fig. S2. [🔗](#)

Video 4. RPE1-based inducible cells expressing GFP-EXOC6A were treated with Dox for 24 h and then serum starved for 30 min. Live cell images were taken using an Elyra 7 SIM Carl Zeiss confocal system under a controlled CO₂ (5%) and temperature (37 °C). Related to Fig. 3C. [🔗](#)

Video 5. RPE1-based inducible cells expressing GFP-EXOC6A were treated with Dox for 24 h and then serum starved for 30 min. Live cell images were taken using an Elyra 7 SIM Carl Zeiss confocal system under a controlled CO₂ (5%) and temperature (37 °C). Related to Fig. 3D. [🔗](#)

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Reviewer #1 (Public review):

Summary:

The study by Lin et al. studies the role of EXOC6A in ciliogenesis and its relationship with the interactor myosin-Va using a range of approaches based on the RPE1 cell line model. They establish its spatio-temporal organization at centrioles, the forming ciliary vesicle and ciliary sheath using ExM, various super-resolution techniques, and EM, including correlative light and electron microscopy. They also perform live imaging analyses and functional studies using RNAi and knockout. They establish a role of EXOC6A together with myosin-Va in Golgi-derived, microtubule- and actin-based vesicle trafficking to and from the ciliary vesicle and sheath membranes. Defects in these functions impair robust ciliary shaft and axoneme formation due to defective transition zone assembly.

Strengths:

The study provides very high-quality data that support the conclusions. In particular, the imaging data is compelling. It also integrates all findings in a model that shows how EXOC6A participates in multiple stages of ciliogenesis and how it cooperates with other factors.

Weaknesses:

The precise role of EXOC6A remains somewhat unclear. While it is described as a component of the exocyst, the authors do not address its molecular functions and whether it indeed works as part of the exocyst complex during ciliogenesis.

<https://doi.org/10.7554/eLife.108271.1.sa2>

Reviewer #2 (Public review):

Summary:

The molecular mechanisms underlying ciliogenesis are not well understood. Previously, work from the same group (Wu et al., 2018) identified myosin-Va as an important protein in transporting preciliary vesicles to the mother vesicles, allowing for initiation of ciliogenesis. The exocyst complex has previously been implicated in ciliogenesis and protein trafficking to cilia. Here, Lin et al. investigate the role of exocyst complex protein EXOC6A in cilia formation. The authors find that EXOC6A localizes to preciliary vesicles, ciliary vesicles, and the ciliary sheath. EXOC6A colocalizes with Myo-Va in the ciliary vesicle and the ciliary sheath, and both proteins are removed from fully assembled cilia. EXOC6A is not required for Myo-Va localization, but Myo-Va and EHD1 are required for EXOC6A to localize in ciliary

vesicles. The authors propose that EXOC6A vesicles continually remodel the cilium: FRAP analysis demonstrates that EXOC6A is a dynamic protein, and live imaging shows that EXOC6A fuses with and buds off from the ciliary membrane. Loss of EXOC6A reduces, but does not eliminate, the number of cilia formed in cells. Any cilia that are still present are structurally abnormal, with either bent morphologies or the absence of some transition zone proteins. Overall, the analyses and imaging are well done, and the conclusions are well supported by the data. The work will be of interest to cell biologists, especially those interested in centrosomes and cilia.

Strengths:

The TEM micrographs are of excellent quality. The quality of the imaging overall is very good, especially considering that these are dynamic processes occurring in a small region of the cell. The data analysis is well done and the quantifications are very helpful. The manuscript is well-written and the final figure is especially helpful in understanding the model.

Weaknesses:

Additional information about the functional and mechanistic roles of EXOC6A would improve the manuscript greatly.

<https://doi.org/10.7554/eLife.108271.1.sa1>

Reviewer #3 (Public review):

Summary:

Lin et al report on the dynamic localization of EXOC6A and Myo-Va at pre-ciliary vesicles, ciliary vesicles, and ciliary sheath membrane during ciliogenesis using three-dimensional structured illumination microscopy and ultrastructure expansion microscopy. The authors further confirm the interaction of EXOC6A and Myo-Va by co-immunoprecipitation experiments and demonstrated the requirement of EHD1 for the EXOC6A-labeled ciliary vesicles formation. Additional experiments using gene-silencing by siRNA and pharmacological tools identified the involvement of dynein-, microtubule-, and actin in the transport mechanism of EXOC6A-labeled vesicles to the centriole, as they have previously reported for Myo-Va. Notably, loss of EXOC6A severely disrupts ciliogenesis, with the majority of cells becoming arrested at the ciliary vesicle (CV) stage, highlighting the involvement of EXOC6A at later stages of ciliogenesis. As the authors observe dynamic EXOC6A-positive vesicle release and fusion with the ciliary sheath, this suggests a role in membrane and potentially membrane protein delivery to the growing cilium past the ciliary vesicle stage. While CEP290 localization at the forming cilium appears normal, the recruitment of other transition zone components, exemplified by several MKS and NPHP module components, was also impaired in EXOC6A-deficient cells.

Strengths:

- (1) By applying different microscopy approaches, the study provides deeper insight into the spatial and temporal localization of EXOC6A and Myo-Va during ciliogenesis.
- (2) The combination of complementary siRNA and pharmacological tools targeting different components strengthens the conclusions.
- (3) This study reveals a new function of EXOC6A in delivering membrane and membrane proteins during ciliogenesis, both to the ciliary vesicle as well as to the ciliary sheath.
- (4) The overall data quality is high. The investigation of EXOC6A at different time points during ciliogenesis is well schematized and explained.

Weaknesses:

(1) Since many conclusions are based on EXOC6A immunostaining, it would strengthen the study to validate antibody specificity by demonstrating the absence of staining in EXOC6A-deficient cells.

(2) While the authors generated an EXOC6A-deficient cell line, off-target effects can be clone-specific. Validating key experiments in a second independent knockout clone or rescuing the phenotype of the existing clone by re-expressing EXOC6A would ensure that the observed phenotypes are due to EXOC6A loss rather than unintended off-target effects.

(3) Some experimental details are lacking from the materials and methods section. No information on how the co-immunoprecipitation experiments have been performed can be found. The concentrations of pharmacological agents should be provided to allow proper interpretation of the results, as higher or lower doses can produce nonspecific effects. For example, the concentrations of ciliobrevin and nocodazole used to treat RPE1 cells are not specified and should be included. More precise settings for the FRAP experiments would help others reproduce the presented data. Some details for the siRNA-based knockdowns, such as incubation times, can only be found in the figure legends.

Taken together, the authors achieved their goal of elucidating the role of EXOC6A in ciliogenesis, demonstrating its involvement in vesicle trafficking and membrane remodeling in both early and late stages of ciliogenesis. Their findings are supported by experimental evidence. This work is likely to have an impact on the field by expanding our understanding of the molecular machinery underlying cilia biogenesis, particularly the coordination between the exocyst complex and cytoskeletal transport systems. The methods and data presented offer valuable tools for dissecting vesicle dynamics and cilium formation, providing a foundation for future research into ciliary dysfunction and related diseases. By connecting vesicle trafficking to structural maturation of an organelle, the study adds important context to the broader description of cellular architecture and organelle biogenesis.

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